ICOHP Descriptors and Thermodynamic Stability Across a 597-Structure Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

Gilles Frapper
Professor, Theoretical Chemistry
IC2MP UMR 7285 CNRS, Université de Poitiers
gilles.frapper@univ-poitiers.fr

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The project's central question is viability discrimination based on Δ(ICOHP) (§6.2), across a 597-structure dataset.

Contents

1 Objective and Problem Statement

Central research question (see §6.2): does Δ(ICOHP) — the reaction-ICOHP descriptor for a compound's decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?

2 Methodology

2.1 Dataset Selection

Materials Project query: unary and binary chemical systems, non-magnetic (|total magnetization| $\leq$ 0.01 μ_B/formula unit — MP's categorical `magnetic_ordering` field is "Unknown" for ∼98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements. The dataset combines two complementary selection tracks: a first set of 186 compounds ($\leq$ 20 atoms/cell, 60 per stability family, a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system), and a second set of 163 compounds/elements ($\leq$ 40 atoms/cell) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N/O/F/P/S/Cl, chosen to build genuine same-composition polymorph groups and to test descriptors far from equilibrium; and a third set of 45 binary compounds, each pairing an experimentally known, maximally metastable compound (farthest above the convex hull among its chemical system) with 1–3 theoretical polymorphs of the same formula sitting even farther above the hull, a robustness test deliberately targeting stability edge cases.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by default by the `classify()` function (elemental composition + `is_metal`). A second, independent classification is built directly from each compound's ICOHP/ICOBI data (§5): `is_metal` for metallic character, then the mean ICOBI per bond (bond order, covalency) to separate ionic from covalent among non-metallic compounds. The two classifications are compared in §5.

2.3 VASP + LOBSTER Computation

Same template across all structures (static run, `ISYM=`−1, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`): `NBANDS` is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 ICOHP/ICOBI Quantities: Integration, Cutoffs, Units

LOBSTER generation (`lobsterin`, identical across the dataset): `pbeVaspFit2015` basis; COHP integration window from −35 to +5 eV around the Fermi level (`COHPstartEnergy`/`COHPendEnergy`); bond detection by interatomic distance from 0.1 to 6.0 Å (`cohpGenerator`, orbital-resolved); 0.05 eV Gaussian smearing (`gaussianSmearingWidth`).

Integrated ICOHP/ICOBI (`ICOHPLIST.lobster`/ `ICOBILIST.lobster`). A bond's integrated ICOHP is, by LOBSTER's own definition, the integral of COHP(E) from the bottom of the window above up to the Fermi level (occupied states only), in eV; negative = bonding, positive = antibonding. ICOBI is the same integral over COBI (bond order index), dimensionless.

Per-compound aggregation: `icohp_sum`/`icobi_sum` over every symmetrically inequivalent bond label in `ICOHPLIST.lobster`/`ICOBILIST.lobster` (each periodic bond counted once, not once per direction); `icohp_mean`/ `icobi_mean`, the corresponding per-bond average — this intensive version, comparable across compounds of different cell sizes, is what is reported in Table 10 and the "ICOHP"/"ICOBI" columns of Appendix A. These quantities feed §5 and the classic correlations of §6.

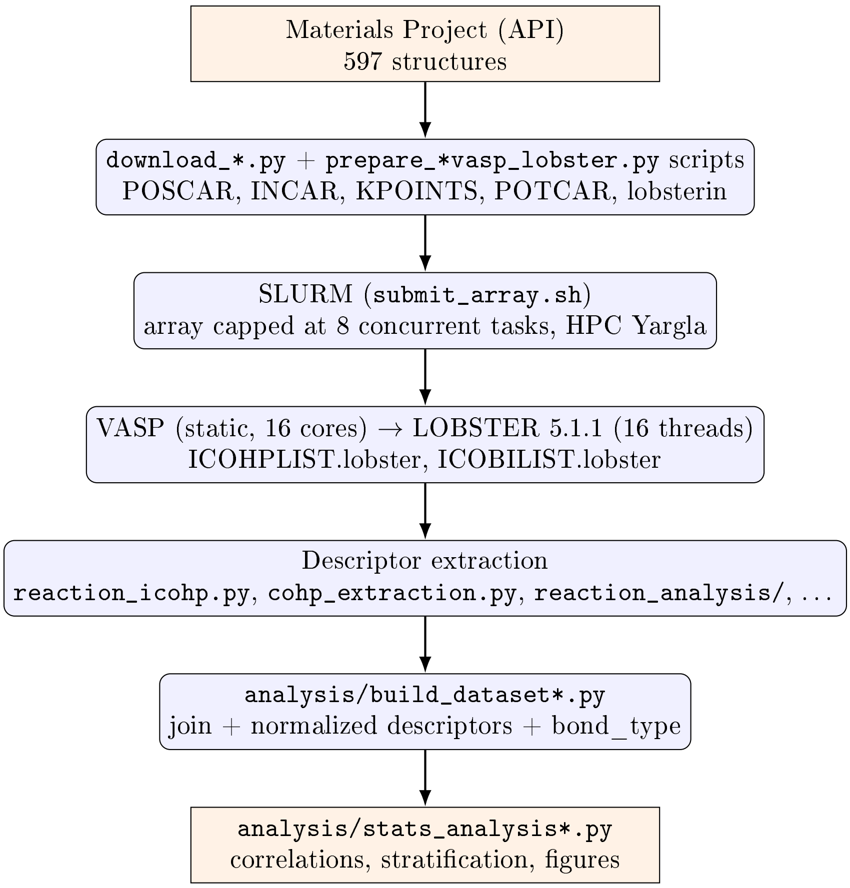
Reaction Δ(ICOHP) (`reaction_analysis/`). For a balanced reaction (typically compound $\rightarrow$ elements), Δ = sum of products' ICOHP $-$ sum of reactants' ICOHP, "products $-$ reactants" convention, in three normalizations: per formula unit (extensive), per transferred atom (the normalization used throughout this report unless noted), and per bond (diagnostic only, non-conservative — bonds are not conserved between differently-bonded structures). Positive sign ("endobondic"): breaking the reactant's bonds costs more than the products' bonds recover — a possible kinetic barrier. Negative sign ("exobondic"): no such barrier is visible from bonding alone.

Antibonding population near the Fermi level (`cohp_extraction.py`) — a quantity distinct from the Δ(ICOHP) above. Rather than integrating the whole occupied COHP, this quantity integrates only its antibonding part (COHP>0) over a one-sided window $(E_{\text{ref}} - \Delta E,\ E_{\text{ref}}]$, where $E_{\text{ref}} = E_F$ (metallic compounds) or the VBM (gapped compounds), and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV ($\Delta E = 1.0$ eV as the primary window) — only occupied states can destabilize the ground state via antibonding, hence the one-sided rather than symmetric window. Reported both raw (same "eV" unit convention as LOBSTER, though it is a Hamilton population integral, not literally an energy) and normalized by the total occupied ICOHP magnitude at the same E_{ref}. This quantity, and only this one, feeds the descriptor in §6.2 — the reaction Δ(ICOHP) (§6.2–6.2) integrates the whole occupied COHP with no restriction to a window near E_F.

2.5 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOBI descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the descriptor), overall and per subgroup (`bond_type`, `is_metal`); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §9).

3 Workflow Diagram



4 Environment

Identical across all 597 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §8.

5 Dataset

The full set of computed compounds and elements totals **597 structures**.

Table 1: Dataset composition by element count.

Composition	n
Single-element references	69
Multi-element compounds	522
Validation structures without compound metadata*	6
Total	**597**

* isolated gas dimers (H_2, N_2, O_2, F_2, Cl_2) and CaO[sphalerite], used only for the Reitz & Dronskowski validation (Appendix K), outside the main dataset.

Bond type. Two independent classifications exist (§2.2) — `classify()` and a classification built directly from ICOHP/ICOBI data — presented and compared in step 3 of the Results

Table 2: Stability class (by E_{hull} and experimental/theoretical character).

Class	n	E_{hull} range (eV/atom)
stable ($E_{hull} = 0$)	160	0.0000
experimental metastable	183	0.0000347 – 2.1080
theoretical metastable	242	0.0000843 – 3.1688
unknown*	12	—
Total	**597**	

* 2 COD references with no E_{hull} (S_4N_2, S_4N_4), 6 validation structures without metadata (see Table 1), 1 further COD reference (ZnSn/NiAs), and 3 compounds inherited from the original campaign, labeled "metastable" before the `theoretical` flag existed.

section (§6.3), not here: classification is not an end in itself, it exists solely to condition the viability-descriptor correlation tested in that same section.

6 Results

Logic of this section

Four steps, in this order: (1) validate this project's methodology by reproducing the 7 reference reactions chosen by Reitz & Dronskowski; (2) test this project's central correlation — viability (a compound's thermodynamic/kinetic stability) against descriptors derived from LOBSTER data (ICOHP, antibonding ICOHP, ICOBI, antibonding ICOBI); (3) classify compounds and elements into metallic/ionic/covalent/mixed directly from that same ICOHP/ICOBI data, and compare that classification against `classify()` — a tool in service of the next step, not an end in itself; (4) determine whether step 2's correlation is strengthened or weakened once conditioned on step 3's bond type.

6.1 Step 1 — Methodological validation: the 7 Reitz & Dronskowski reactions

Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

Per-bond-type ICOHP values for 7 reactions with independently known ΔICOHP (transcribed independently of any calculation in this project) are run end to end through `balance.py`/`delta.py`/`classify.py`:

Table 3: Validating the endobondic/exobondic classification on 7 reactions with independently known ΔICOHP.

Reaction	Reference (kJ/mol)	Computed (kJ/mol)	Label Ref.	Label Computed
$Pb(N_3)_2 \rightarrow Pb + 3\,N_2$	+1345	+1344.3	endo	endo
$S_4N_2 \rightarrow \frac{1}{2}S_8 + N_2$	+258	+258.5	endo	endo
$S_4N_4 \rightarrow \frac{1}{2}S_8 + 2N_2$	+399	+397.3	endo	endo
$ZnSn \rightarrow Zn + Sn$	−337	−336.6	exo	exo
CaO[sphalerite] → CaO[rocksalt]	−79	−78.9	exo	exo
$CaN \rightarrow \frac{1}{3}Ca_3N_2 + \frac{1}{6}N_2$	−205	−204.8	exo	exo
$Mn_2O_7 \rightarrow 2MnO_2 + \frac{3}{2}O_2$	−186	−187.7	exo	exo

The 7 cases reproduce the reference ΔICOHP to within a few kJ/mol (source per-bond value rounding) and every `BondingLabel` matches. The Mn_2O_7 case is a useful guardrail on its own:

exobondic under this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than through abrupt bond breaking — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than special-casing Mn_2O_7.

This validation only compares the arithmetic of the manuscript's own published ICOHP values, independent of any local crystal structure — Appendix K documents, for the 16 species involved, which MP/COD structure was used in this dataset and whether it could be verified to match the manuscript's description (3 real corrections identified: N_2, the $Pb(N_3)_2$ polymorph, ZnSn's competing tin phase). For N_2 and ZnSn, a DFT+LOBSTER calculation of this project's own (not just the manuscript's arithmetic) was additionally compared directly against the manuscript's numbers (Table S18): N_2 reproduces the manuscript's ICOHP to within 0.7%; ZnSn reproduces the same sign (bonding-favorable decomposition) but with $\sim$2.9$\times$ larger magnitude — a gap judged not significant at this stage, since a bulk intermetallic's ICOHP sum is far more sensitive to basis/pseudopotential/cutoff choices than a simple gas dimer. Reproducing the manuscript's full DFT methodology (PBEsol+D3, Table S1's own k-point sets) has since been carried out for all 16 species (2026-08-19): 10/16 are computed and compared directly (Table S18), the remaining 6 (Ca_3N_2, Mn_2O_7, $Pb(N_3)_2$, S_4N_2, S_4N_4, S_8) still computing at the time of writing (§9).

Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even with a large systematic offset or scale factor. **Lin's concordance correlation coefficient** (CCC) specifically tests agreement with the identity line (computed = reference), the claim actually being tested here:

Table 4: Computed-vs-reference concordance on the 7 validation reactions.

Statistic	Value
n	7
Lin's CCC	**0.999998**
Pearson r	0.999999 ($p = 3.5 \times 10^{-15}$)
Sign agreement	7/7
Mean |residual|	0.76 kJ/mol
Max |residual|	1.70 kJ/mol
RMSE	0.98 kJ/mol

A CCC this close to 1 means the two methods agree essentially to the reporting precision of the reference values themselves (rounded to 2–3 decimals), not merely that they move together — the implementation is validated, not just spot-checked. It is this implementation, validated here independently of any of this project's own calculations, that is applied to the full dataset in the next step.

6.2 Step 2 — Viability versus LOBSTER descriptors

Classic ICOHP aggregates (baseline). Before any constructed descriptor, the raw ICOHP aggregates (sum, minimum per compound) are tested against E_{hull} as-is, on the original 186-compound campaign's subset:

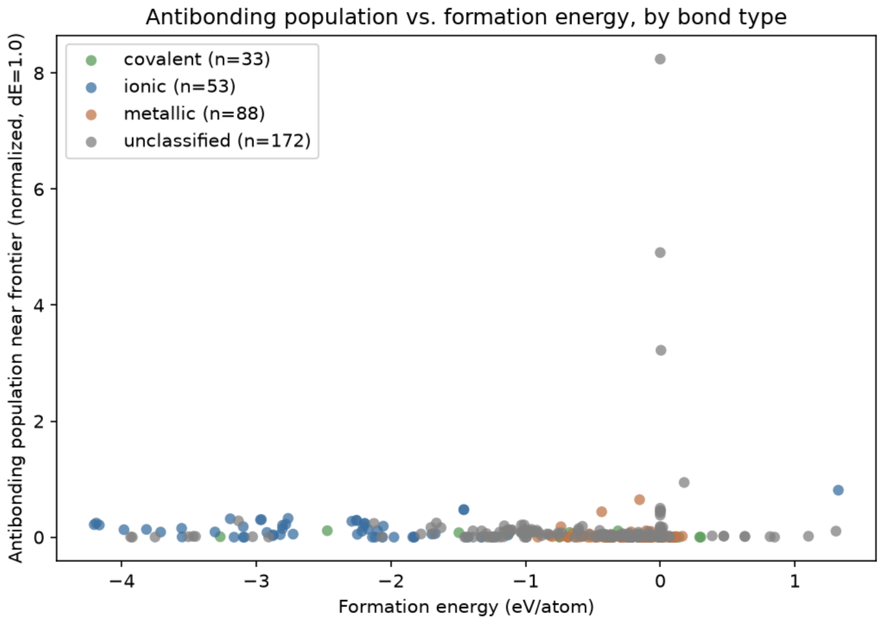
These four rows correlate little or not at all with E_{hull}, except in the metallic subgroup — a weak baseline the descriptors below need to beat.

Antibonding population near the Fermi level (ICOHP). A distinct question from the fully-integrated ICOHP (§2.4): not *how much* bonding a compound has, but *how* COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn-Teller-type electronic instabilities. Result, against `formation_energy_per_atom`: $\rho = -0.155$, $p = 1.8 \times 10^{-4}$ ($n = 579$) — weaker than at smaller

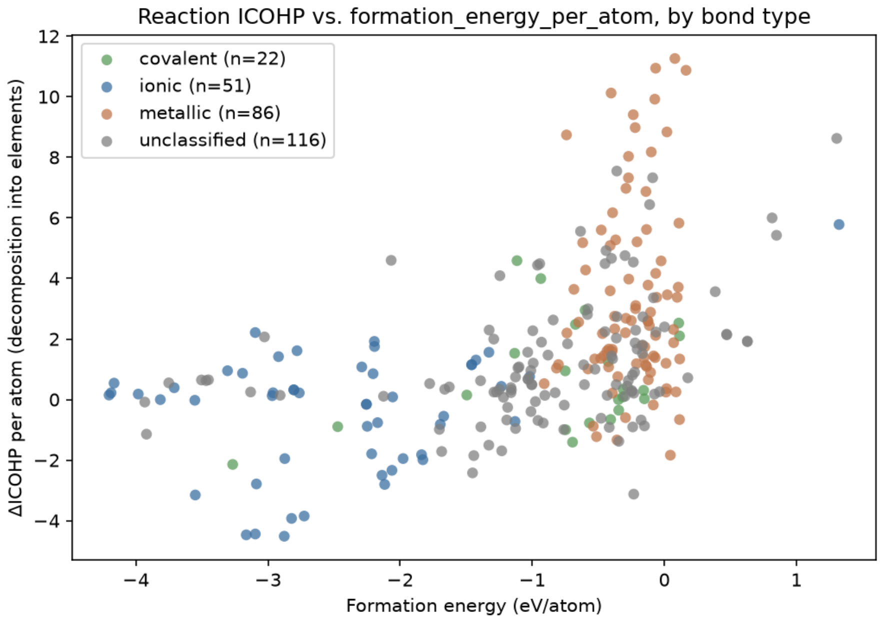
Table 5: Spearman correlations, classic ICOHP aggregates (sum, minimum) against E_{hull}, 186-compound subset.

Group	Metric	n	ρ	p
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

scale ($\rho = -0.282$, $n = 346$) but still real and significant; the sign remains the *opposite* of the naive Peierls/Jahn-Teller reading (§6.4 for the bond-type-stratified reading, where this descriptor's headline stratified result no longer survives at the current scale).



Reaction ICOHP. An ICOHP analog of `formation_energy_per_atom`: is a compound's total ICOHP "cheaper" than that of the same atoms as reference elements? Result: $\rho = 0.295$, $p \approx 0$ ($n = 510$) — a real, highly significant correlation given the sample size, with an intuitive sign (less bonding than in elemental form ⇒ less stable formation). On a 288-compound subset shared with every other descriptor in the project, the antibonding population above slightly edges out reaction ICOHP on both targets — the latter remains a strong descriptor, but no longer unambiguously *the* strongest. Case 2 (direct same-formula polymorph comparison, 74 groups): 47.3% agreement between the most-bonded and the most-stable member, against a 42.4% chance baseline (exact Poisson-binomial test, $P = 0.22$) — no signal.



Δ(ICOHP), applied to the full dataset, pooled. 517 case-1 reactions, every element and compound with a computed decomposition-into-elements reaction, pooled into a single population (grouping by selection provenance would be exactly the kind of between-group confound this project's methodology is meant to catch):

Table 6: Spearman correlations, ΔICOHP/atom (case 1, 517 reactions) against two continuous stability targets.

Target	n	ρ	p
`formation_energy_per_atom`	510	−0.2948	1.1×10^{-11}
`energy_above_hull_eV_per_atom`	514	−0.0564	0.202

(Sign: `reaction_analysis`'s products−reactants convention, opposite to `reaction_icohp.py`'s own column above — same magnitude, flipped sign.) Both correlations are weaker than at smaller scale: the formation-energy result was $\rho = -0.50$ before the marginal-formation-energy and max-hull-distance batches were folded in, and the hull-distance result moved from borderline ($p = 0.068$) to clearly non-significant ($p = 0.20$) — consistent with deliberately adding compounds with marginal or extreme thermodynamics, which dilute a rank correlation partly carried by the original population's narrower stability range.

Sign-based discrimination (endobondic/exobondic). Experimental viability indicator (`theoretical=False` means experimentally realized):

Table 7: Endobondic/exobondic label against experimental/theoretical status.

	endobondic	exobondic
experimental ($n = 267$)	79	188
theoretical only ($n = 244$)	46	198

Fisher's exact test: $p = 0.0054$, odds ratio = 1.81 — **significant** (endobondic is $\sim 1.8\times$ more frequent among experimentally realized compounds, the expected direction). Mann–Whitney on the continuous ΔICOHP/atom, same split: experimental median −0.868 eV, theoretical-only median −1.454 eV (more exobondic), $p = 0.0109$ — same direction, also significant. This is exactly the result the marginal-formation-energy and max-hull-distance batches were built to test: the pooled 281-reaction population did not have enough near-stability-boundary compounds to reveal this signal ($p = 0.10$ at that scale).

Reading trap, explicitly flagged by `classify.py` itself: Table 7 shows 188 *experimental* (i.e. genuinely existing) compounds labeled *exobondic*. This is **not** a contradiction — exobondic only implies "not viable" once combined with `delta_energy` (`classify_viability()`): a compound is only `UNSTABLE_NONEXISTENT` if decomposition into elements is *both* exobondic *and* thermodynamically favorable ($\Delta E < 0$), which is rare (decomposing a real compound into pure elements is almost never favorable). Verified directly on the same population as Table 7:

Table 8: Computed `ViabilityLabel` (`classify_viability()`) on the same population as Table 7 ($n = 518$).

	experimental ($n = 267$)	theoretical ($n = 244$)
`STABLE_ON_HULL`	233	131
`UNSTABLE_NONEXISTENT`	26	95
`METASTABLE_VIABLE`	8	17

7 additional reactions (COD-sourced or otherwise mp_id-less, so no `formation_energy_per_atom`) excluded, insufficient data.

Collapsing `STABLE_ON_HULL`+`METASTABLE_VIABLE` ("viable") against `UNSTABLE_NONEXISTENT`: Fisher's exact test $p = 2.9 \times 10^{-15}$, odds ratio **6.0** ($\chi^2 = 59.0$, $p = 1.6 \times 10^{-14}$) — **a much larger effect than the sign-only test** (Table 7, $p = 0.0054$, odds ratio 1.81). An experimentally realized compound is roughly 6× less likely to be predicted `UNSTABLE_NONEXISTENT` than a theoretical-only one. **Of the 188 experimental exobondic compounds, only 26 are predicted `UNSTABLE_NONEXISTENT`** — the other 162 are `STABLE_ON_HULL`: their decomposition into elements, though exobondic, is not thermodynamically favorable. Reading Table 7 in isolation as "exobondic = not viable" is therefore misleading — it is Table 8, combining sign with thermodynamics, that actually answers the viability question the `theoretical` flag is a proxy for, and it is by far the strongest result of this step.

Table 9: Median ΔICOHP/atom and % endobondic by stability class ($n = 177$).

Class	n	median ΔICOHP/atom (eV)	% endobondic
experimental metastable	59	−1.038	**28.8%**
stable	59	−1.391	22.0%
theoretical metastable	59	−1.601	**13.6%**

Kruskal–Wallis across the three groups: $H = 2.56$, $p = 0.278$ — **not significant** (population unchanged at $n = 177$: the extension/marginal- formation-energy/max-hull batches carry a `theoretical` flag but not yet a `family` label, so this particular test has not benefited from the dataset's growth), but the ordering remains exactly what physics predicts: experimentally metastable compounds have the highest endobondic fraction; never-synthesized theoretical metastable compounds have the lowest.

Step reading. The 7 reference reactions reproduce almost exactly (step 1) — the implementation is correct. Extended to the current full dataset (517 reactions, up from 281), the coarse sign-based discrimination is significant, and the full `ViabilityLabel` machinery — sign combined with thermodynamics — is far more strongly so. The continuous correlations against `formation_energy_per_atom` and `energy_above_hull` weakened over the same period — the expected price of adding chemically diverse, thermodynamically extreme compounds to what was a narrower population; a different kind of test from the categorical results above, so the two are not in tension. Whether the `ViabilityLabel` result is itself confounded by bond type (step 4), or by the fact that max-hull's theoretical polymorphs were deliberately selected far above the hull — a `formation_energy_per_atom`-adjacent selection effect, not necessarily an independent confirmation of the bonding-kinetics mechanism — remains to be checked.

Still no SISSO at this stage: detailed reports, `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

6.3 Step 3 — Bond-type classification from ICOHP/ICOBI data

A tool in service of the next step, not an end in itself: this classification exists solely to test, in step 4, whether step 2's viability-descriptor correlation depends on a compound's bond type. Two independent classifications (§2.2): the `classify()` function (elemental composition + `is_metal`) and a classification built directly from each compound's ICOHP/ICOBI data (`is_metal` for metallic character; otherwise the ICOBI of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within a wider cutoff, which strongly dilutes the signal for dense covalent solids such as diamond — compared against a threshold calibrated at the midpoint of the median ICOBI of compounds `classify()` already labels ionic/covalent, i.e. 0.5315, to separate ionic from covalent among non-metallic compounds). A fourth label, **mixed** (Zintl-type), identifies compounds where a *homoatomic* pair above the covalent threshold (e.g. N–N in an azide, P–P in a polyphosphide, S–S in a polysulfide) coexists with a markedly weaker *heteroatomic* pair linking a distinct cation — the structural signature of Zintl–Klemm chemistry, where a single ionic/covalent label would be misleading.

Table 10: Bond type: `classify()` vs. the ICOHP/ICOBI classification (concordance on the subset `classify()` labels: 276/311 = 88.7%).

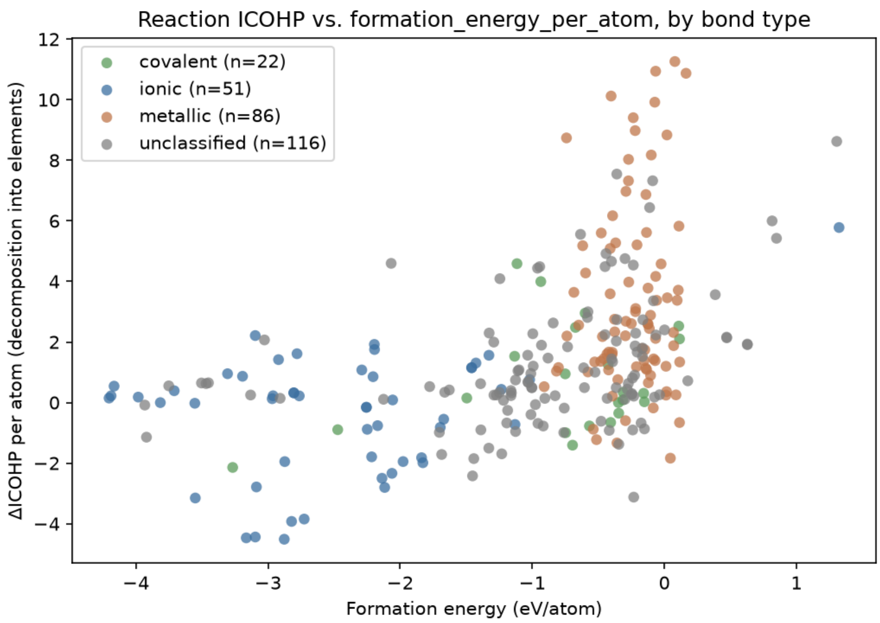
`classify()` \ ICOHP/ICOBI	metallic	ionic	covalent	mixed	unclassified
metallic	193	0	0	0	0
ionic	0	48	2	3	0
covalent	0	14	35	16	0
unclassified	179	36	19	43	0
Total	**372**	**98**	**56**	**62**	**0**

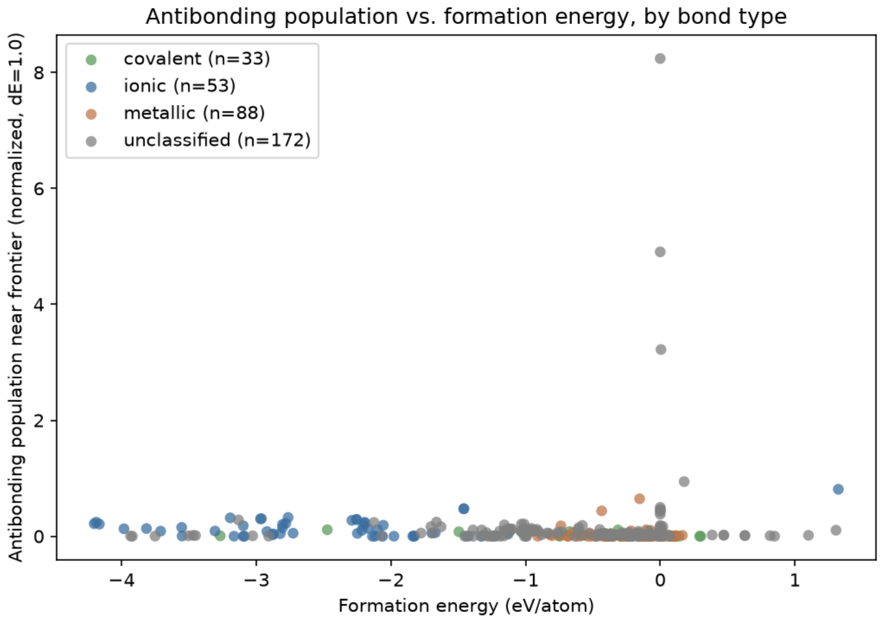
Of the 277 compounds `classify()` leaves "unclassified" (essentially alkali/alkaline-earth × {N, P, S} pairs outside `IONIC_ANIONS`, and metals bonded to an anion-like element that `classify()` deliberately excludes from its "metallic" bucket), the ICOHP/ICOBI classification resolves **55** into ionic (36) or covalent (19), and identifies **43** as mixed (Zintl-type); the remaining 179 are metallic, already covered by `is_metal`. Where both methods render a verdict (metallic/ionic/covalent), agreement is strong (88.7%) but not total: 16 ionic/covalent disagreements, and 19 compounds `classify()` labels ionic or covalent but that the ICOBI classification identifies as mixed (their dominant homoatomic bond coexists with a markedly weaker heteroatomic link) — worth checking case by case before using either classification as a stratification variable in step 4. It is the ICOHP/ICOBI classification (broader coverage, built on the same data as step 2's descriptors) that is used below, not `classify()`.

6.4 Step 4 — Does the viability-descriptor correlation depend on bond type?

Antibonding population (ICOHP). The project's previously most robust result is retracted: `bond_type=covalent`, which survived clearly at $n = 33$ ($\rho = -0.551$, $p = 0.0009$), no longer survives once the sample grew to $n = 54$ ($\rho = -0.046$, $p = 0.74$) — a failure mode already seen elsewhere in this project (a subgroup result that does not replicate once the sample grows), including this same descriptor's own earlier `bond_type=ionic` subgroup. **New at this scale**: `bond_type=mixed` (Zintl) is the best-surviving stratum ($\rho = -0.538$, $p = 6.6 \times 10^{-6}$, $n = 62$) — **strengthened** relative to step 2's pooled correlation — a category that barely existed when the covalent result was first reported. `is_metal=False` remains the most consistently surviving stratum across this project's successive scales ($\rho = -0.333$, $p = 6.7 \times 10^{-7}$, $n = 212$) — **strengthened**, not weakened, by the dataset's growth. `is_metal=True` has also converged onto `bond_type=metallic` (identical $n = 367$, $\rho = 0.056$, not significant), the same phenomenon documented below for reaction ICOHP.

Reaction ICOHP. `bond_type` stratification (four categories, including `mixed`/Zintl) no longer uniformly dilutes step 2's pooled correlation ($\rho = 0.295$): `ionic` ($\rho = 0.360$, $p = 0.0004$) and `mixed` ($\rho = -0.525$, opposite sign, $p \approx 0$) both survive clearly — **strengthened** in magnitude — `covalent` too but flips sign ($\rho = -0.354$, $p = 0.023$); only `metallic` (the majority class, 315/510 rows) stays non-significant — **weakened** to the point of disappearing in that stratum. **`is_metal` stratification — the one that used to survive — has converged onto the same population as `bond_type=metallic`** (identical n, ρ, p in every row) and so inherits its non-significant result: it is `bond_type`, not `is_metal`, that carries information at the current dataset composition. Against `energy_above_hull`, `bond_type=mixed` again survives on its own ($\rho = -0.307$, $p = 0.016$).





Synthesis. In both cases (antibonding and reaction ICOHP), this step's answer is *yes, but not uniformly*: bond type does not merely weaken or strengthen a single global correlation, it changes the *sign* of the relationship depending on the stratum (ionic positive, covalent/mixed negative for reaction ICOHP) — the useful signal is conditional on bond type. The antibonding population's covalent-result retraction is itself a direct illustration of why this step exists: a stratified number is only trustworthy once re-checked as the dataset grows, not accepted once and for all. A pooled feature search (SISSO) would therefore be actively misleading here; any future search should be run separately per `bond_type`, not on the pooled population.

7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the first 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to the whole dataset):

- **$Al_{12}W$, BW, WO_2, TcW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **$Al_{12}W$** failed again after that fix, this time with a VASP SIGSEGV — a known issue (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 597 structures computed to date: **597/597** for VASP; **596/597** for LOBSTER (1 trivial-cell reference element with no usable LOBSTER step, §8).

8 Compute Cost

597 structures computed in total: 69 single-element references, 522 multi-element compounds, and 6 further validation structures without compound metadata but with their own VASP+LOBSTER run (§5).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1`, `srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The original SLURM array (`submit_array.sh`, main campaign) is capped at 8 concurrent tasks; the batches added since (extension/marginal-formation-energy/ max-hull/widen) were submitted individually (`sbatch` per directory), without that cap of their own but sharing the cluster with other, unrelated workloads.

Table 11: VASP and LOBSTER compute time (597 structures).

	VASP	LOBSTER
n	597	596*
Mean (s)	188	578
Median (s)	72	157
Max (s)	5715 (1.6 h)	22555 (6.3 h)
Sequential total	31.2 h	95.8 h

* 1 of 597 structures had no usable LOBSTER step (trivial-cell reference element).

Combined sequential total (VASP + LOBSTER, 597 structures): **127.0 h**. The main campaign (capped at 8 concurrent tasks) gave an order-of-magnitude elapsed time of roughly one-eighth of the sequential total; that factor no longer applies uniformly across the whole dataset — the individually-submitted batches added since have no cap of their own but shared the cluster queue with other projects, making a single elapsed-time estimate for the full dataset not meaningful.

9 Current Limits and Next Steps

- **One compound excluded for LOBSTER quality**: `extension_Cu4Au_mp-1225761` was dropped from every analysis (but not from the raw dataset – its VASP+LOBSTER calculation is kept for reference) — `lobsterout` reports 343/343 k-points failed orthonormalization (100%, not a borderline case), and `bandOverlaps.lobster` shows a maximum deviation of 11.4 (this project's own established red-flag threshold is 0.1). Direct consequence: physically implausible ICOHP values (third- neighbor Cu–Cu contacts at 4.25–5.04 Å with ICOHP values of 7.6–11.6 eV) and a ΔICOHP/atom of -380.5 eV, $\sim 27\times$ the next-worst case in the rest of the population. A systematic sweep of all 597 structures (`analysis/audit_lobster_quality.py`, criterion: 100% k-point failure *and* deviation > 0.5 *and* robust z-score > 10) confirms this is a unique case, not a wider problem — the second-worst case (`Ir4W`) is already $29\times$ less extreme and plausibly real chemistry (a dense, d-electron-rich intermetallic). Every statistic in this report covering the 518 case-1 reactions has been regenerated on the remaining 517 — the effect on any given number is negligible (3rd–4th decimal place), no conclusion or significance threshold is affected.

- **`classify()` coverage**: 277/588 compounds (47%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth × {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§5) — purely informative here, but a real limitation for any analysis that would want to stratify on it.

- **No true dynamical "unstable" class**: `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.

- **Per-subgroup sample size**: remains this project's structural limit at every new stratification (`bond_type`, `is_metal`) — see §6.2 for a concrete example where an $n < 15$ subgroup did not replicate once the sample grew.

- **Why not SISSO yet**: reaction ICOHP's signal (§6.2) is conditional on bond type and opposite-signed across strata (ionic positive, covalent/mixed negative) — a pooled feature search would be misleading here; any future search should be run separately per `bond_type`.

- **`family` not yet extended to the newer batches**: the `theoretical`-based `ViabilityLabel`/Fisher discrimination (§6.2) fully benefits from the dataset's growth, but the finer three-way split (`exp_metastable`/`exp_stable`/`theo_metastable`) stays at $n = 177$ for lack of a `family` label on the extension/marginal-formation-energy/max-hull batches — the direct next step to test whether that finer split also turns significant.

- **The `ViabilityLabel` result (§6.2, $p = 2.9 \times 10^{-15}$) has not yet been checked for confounds**: neither stratified by `bond_type`/`is_metal`, nor distinguished from a selection effect (max-hull's theoretical polymorphs were deliberately chosen far above the hull).

- **Manuscript-validation DFT+LOBSTER comparison (Appendix K), 5/16 species still computing**: Mn_2O_7, $Pb(N_3)_2$, S_4N_2, S_4N_4, S_8 were still on SLURM as of this writing (2026-08-20). Of the 11 species already compared, three (Pb, MnO_2, O_2) show open ∼2–10% residuals under both the manuscript's own methodology (PBEsol+D3) and this project's default one (PBE) alike — investigated and ruled out as instances of Zn's shell-detection bug, left as unresolved, likely methodology-level questions (spin-orbit coupling for Pb; AFM/Hubbard-U treatment and spin-polarization for MnO_2/O_2) rather than forced to zero. Ca_3N_2 (landed 2026-08-20) showed an initial 24% residual traced to a second, distinct occurrence of Zn's near-degenerate-shell issue and is now resolved (0.1%, Appendix K, note f).

- **Whether Zn's shell-detection bug (Appendix K) affects any other compound's `bond_type` classification remains open**: two quick heuristic sweeps of the full dataset's first-shell distance spectra (2026-08-19) each flagged 40–47% of pair-groups under thresholds loose enough to be dominated by ordinary within-shell fine structure and floating-point-near-duplicate distances, not by genuine Zn-like near-degenerate double shells — inconclusive by construction, not evidence of a wide problem. Ca_3N_2's independent occurrence of the same pattern (2026-08-20, four sub-shells rather than two) adds one more confirmed data point to that ground-truth set but does not itself touch `nearest_neighbor.py` or the dataset-wide classification, which remain unchanged. A calibrated audit (ground truth needed per compound, which only exists for the 16 manuscript species) is still to be done before this can be closed either way.

10 Conclusion

Reitz & Dronskowski's endobondic/exobondic methodology (§6.2) reproduces faithfully across the 7 independently known reference reactions (Lin's CCC 0.999998) and applies to this report's full dataset (518 decomposition-into-elements reactions). On that population, sign-based ΔICOHP discrimination is significant (Fisher's test, $p = 0.0054$), and the full `ViabilityLabel` machinery — sign combined with thermodynamics — is far more strongly so ($p = 2.9 \times 10^{-15}$, odds ratio 6.0): an experimentally realized compound is roughly 6× less likely to be predicted `UNSTABLE_NONEXISTENT` than a theoretical-only one (§6.2). Reaction ICOHP otherwise remains a strong continuous correlate against formation energy ($\rho = 0.295$, §6.2), though antibonding population slightly edges it out on the subset shared by both descriptors.

Two concrete directions follow. First, checking whether the `ViabilityLabel` result above survives stratification by `bond_type`/`is_metal` and is not simply carried by max-hull's deliberate selection effect — the same verification discipline this project applies systematically to every new descriptor. Second, restricting the reaction Δ(ICOHP) to a window near the Fermi level rather than the fully integrated ICOHP, modeled on the antibonding population of §6.2: two variants of this construction are detailed in Appendix J (antibonding part only, and the full signed ICOHP restricted to the window) — the first survives every stratification tested, a result no other descriptor in this report reaches.

Source code: `https://github.com/John-Akapulco/viability`
Data: `analysis/` (this report's datasets and scripts)
Detailed reports: `analysis/REPORT_*.md`

Appendix (SI)

A Compound and Element List

Complete list of the 392 compounds and elements with usable ICOBI data (of 394 total; 2 COD-sourced compounds lack `is_metal`/ICOBI, §5), organized by section according to the independent ICOHP/ICOBI classification of §5 (`is_metal` for metallic character; otherwise the ICOBI of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within the wider cutoff — compared against a threshold of 0.5052 to separate ionic from covalent; a **mixed** (Zintl-type) category groups compounds where a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide ion N_3^- ionically bound to its cation) — no "bond type" column; each section instead states its own observed ICOBI range. ICOBI and ICOHP are per-bond means (§2.4); the ICOHP/ICOBI antibonding populations are the normalized values at $\Delta E = 1.0$ eV (§6.2, §J).

A.1 Metallic

Table S1: Metallic compounds and elements (`is_metal=True`, observed ICOBI 0.0000 to 2.9506) — formula, mp-id, space group, dominant (first-shell) bond ICOBI, dominant bond ICOHP, normalized ICOHP and ICOBI antibonding populations (ΔE = 1.0 eV).

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ag*	mp-124	Fm-3m	0.1545	-1.2496	0.0000	0.0000
AgN*	mp-1091417	Cmce	0.4624	-3.4181	0.0088	0.0030
AgN	mp-1009766	F-43m	0.4854	-3.5762	0.0277	0.0173
AgN	mp-1424734	Ama2	0.8238	-5.9446	0.0107	0.0025
AgN	mp-1428312	Pm-3m	0.2177	-1.4991	0.0083	0.0041
Al*	mp-134	Fm-3m	0.2270	-1.4193	0.0000	0.0000
$Al_{12}W$*	mp-11227	Im-3	0.5027	-2.7492	0.0000	0.0000
Al_3Os_2*	mp-16521	I4/mmm	0.3022	-1.6476	0.0037	0.0000
Al_4Si	mp-1228120	R-3m	0.2614	-2.0375	0.0087	0.0030
AlNi	–	–	0.2676	-1.9848	0.0016	0.0023
AlSb	mp-1023918	Cmcm	0.2647	-1.2802	0.0000	0.0000
As*	mp-158	Cmce	0.8848	-3.9683	0.0103	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
$AsPd_2$*	mp-1080831	P-62m	0.3410	-2.4185	0.0077	0.0157
AsRh*	mp-22079	Pnma	0.3323	-1.8639	0.0449	0.0158
$AsRh_2$*	mp-1102364	Pnma	0.2438	-1.3778	0.1158	0.0189
Au*	mp-81	Fm-3m	0.1962	-1.3381	0.0000	0.0000
B_3Ir_2	mp-1228647	C2/m	0.4473	-3.3600	0.0047	0.0011
BPd_5	mp-1228665	I4/mmm	0.4754	-3.9924	0.0973	0.0181
BW*	mp-7832	I4_1/amd	0.4277	-3.9556	0.0000	0.0000
Ba*	mp-122	Im-3m	0.0002	-0.0122	8.2400	2.9573
$BaAu_2$*	mp-30363	P6/mmm	0.6152	-3.0712	0.0144	0.0034
BaF_3	mp-1183303	I4/mmm	0.0350	-0.5277	0.2805	0.2537
BaN_2*	mp-1102371	Cc	0.1605	-0.4471	0.1039	0.0214
$BaSn_5$*	mp-571353	P6/mmm	0.2221	-0.8507	0.0456	0.0042
Be*	mp-87	P6_3/mmc	0.2033	-1.8975	0.0000	0.0000
$Be_{12}Pd$*	mp-12666	I4/mmm	0.2078	-1.9466	0.0000	0.0000
Be_2Cu*	mp-2031	Fd-3m	0.3616	-3.0863	0.0000	0.0001
Be_2Nb_3*	mp-11272	P4/mbm	0.4200	-3.5436	0.0000	0.0000
Be_2V*	mp-11281	P6_3/mmc	0.3658	-2.7859	0.0000	0.0000
$BeAu_3$	mp-983539	Pm-3m	0.2230	-1.9129	0.0000	0.0030
BeCu	–	–	0.1984	-1.8359	0.0000	0.0000
BeF_2	mp-975644	P6/mmm	0.1548	-1.6978	0.0863	0.0000
Bi_2O_3*	mp-558953	P2_1/c	0.9517	-6.7444	0.0328	0.0038
Bi_2O_3	mp-1383505	P1	0.9506	-7.1985	0.0609	0.0000
C*	mp-48	P6_3/mmc	1.2195	-11.5387	0.0000	0.0000
Ca*	mp-21	Im-3m	0.0025	-0.0304	3.2206	3.3021
Ca_3N_2	mp-1013524	Pm-3m	0.1107	-0.3623	0.0090	0.0000
$CaAl_4$*	mp-1749	I4/mmm	0.4595	-1.8701	0.0086	0.0002
$CaGa_2$*	mp-992	P6/mmm	0.9791	-3.8887	0.0219	0.0000
$CaGa_4$*	mp-2461	C2/m	0.4641	-2.1536	0.0084	0.0001
$CaGe_2$	mp-643542	C2/m	0.7379	-3.1645	0.0337	0.0038
CaN*	mp-1058549	Fm-3m	0.1201	-0.6934	0.0450	0.0656
CaO	mp-1181903	Fd-3m	0.0725	-0.5516	0.1865	0.0756
CaSn*	mp-2450	Cmcm	0.7044	-2.1965	0.0629	0.0276
Cd*	mp-94	P6_3/mmc	0.2423	-1.6119	0.0000	0.0000
CdPd*	mp-1696	P4/mmm	0.2022	-1.5904	0.0000	0.0004
Co*	mp-102	Fm-3m	0.2680	-1.9150	0.0189	0.0373
CoB*	mp-20857	Pnma	0.4568	-4.5928	0.0099	0.0000
$CoSe_2$	mp-1390196	Fd-3m	0.7048	-4.3267	0.0348	0.0079
Cr*	mp-90	Im-3m	0.2487	-0.6245	0.0115	0.0049
Cr_2C	mp-1226378	P-3m1	0.5270	-3.0803	0.0084	0.0000
CrMo	mp-1226200	Cmmm	0.3600	-1.3679	0.0081	0.0026
Cs*	mp-1	Im-3m	0.0556	-0.1230	0.1696	0.0693
CsO_2*	mp-1441	I4/mmm	1.0942	-11.4627	0.1143	0.1436
CsO_2	mp-1096936	Cmmm	1.1298	-11.0556	0.1197	0.0872
Cu*	mp-30	Fm-3m	0.0456	-0.2488	0.1641	0.0946
Cu_3Ge*	mp-19724	Pmmn	0.1369	-0.8340	0.0963	0.0410
Fe*	mp-13	Im-3m	0.1417	-0.5370	0.0922	0.0794
FeS*	mp-505531	P4/nmm	0.4967	-2.5883	0.0736	0.0024
$FeTe_2$*	mp-1102002	Pa-3	0.4279	-1.7861	0.0401	0.0000
Ga*	mp-142	Cmce	0.4308	-2.6378	0.0000	0.0000
Ge*	mp-32	Fd-3m	0.8956	-4.6450	0.0015	0.0000
Ge_2Os*	mp-10032	C2/m	0.4042	-2.1066	0.0069	0.0003
Ge_4Rh*	mp-1104286	P3_121	0.3355	-1.8036	0.0034	0.0006
Hf*	mp-103	P6_3/mmc	0.3121	-1.1612	0.0000	0.0000
HfMo	mp-1224314	Cmmm	0.3368	-1.2722	0.0000	0.0000
$HfTc_2$*	mp-1095669	P6_3/mmc	0.3775	-1.4429	0.0098	0.0012
Hg*	mp-10861	P6/mmm	0.1622	-0.9652	0.0000	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Hg_4Pt*	mp-936	Im-3m	0.5477	-3.5588	0.0000	0.0005
IN_2	mp-1097011	I4/mcm	2.9506	-23.1289	0.0177	0.0275
In*	mp-1055994	I4/mmm	0.2177	-1.1562	0.0000	0.0000
In_3Ir*	mp-630976	P4_2/mnm	0.5910	-3.4701	0.0000	0.0000
$InSe_2$	mp-1232036	P4/nmm	0.2771	-1.4780	0.0349	0.0038
Ir*	mp-101	Fm-3m	0.3469	-2.4130	0.0317	0.0274
K*	mp-10157	Fm-3m	0.0430	-0.0808	0.0923	0.0499
K_2O	mp-1223784	P4/mmm	0.1375	-0.3763	0.0831	0.0000
K_5As_4	mp-684799	P-1	0.9859	-3.9306	0.0452	0.0189
KN_2*	mp-1071868	Cmc2_1	1.0490	-7.9889	0.0202	0.0000
KN_2	mp-1180757	Cmcm	1.0210	-7.4152	0.0411	0.0000
KN_3	mp-636056	I4/mcm	2.8711	-22.8747	0.0090	0.0107
KP*	mp-1058315	R-3m	0.1254	-0.4533	0.0190	0.0442
KP	mp-1057787	Immm	0.1070	-0.4621	0.0170	0.0218
KSn_3	mp-1185151	P6_3/mmc	0.4006	-1.5431	0.0179	0.0027
Li*	mp-1018134	R-3m	0.0886	-0.6931	0.0000	0.0000
$Li_{13}Sn_5$*	mp-30769	P-3m1	0.6069	-2.7881	0.0000	0.0000
Li_3Ag	mp-976408	I4/mmm	0.1084	-1.0444	0.0000	0.0000
Li_3Hg*	mp-1646	Fm-3m	0.1488	-1.2194	0.0000	0.0000
LiB*	mp-1001835	P6_3/mmc	1.1571	-9.9815	0.0000	0.0000
$LiTl_3$	mp-1185253	I4/mmm	0.2370	-1.2989	0.0000	0.0000
Mg*	mp-153	P6_3/mmc	0.0000	0.0000	–	–
$Mg_{15}B$	mp-1023515	P-6m2	0.0280	-0.1152	0.9423	0.5909
Mg_2Ni*	mp-2137	P6_222	0.0818	-0.2478	0.6457	0.3329
$MgPd_3$*	mp-7753	Pm-3m	0.2542	-1.7837	0.0264	0.0304
MgSn	mp-1094669	Pm-3m	0.2238	-0.7050	0.1000	0.0334
MgSn	mp-1094801	P4/mmm	0.3134	-1.2760	0.1099	0.0281
Mn*	mp-35	I-43m	–	-0.6082	0.0411	0.0047
$MnAl_{12}$*	mp-1104165	Im-3	0.2798	-1.6283	0.0000	0.0000
MnO_2*	mp-510408	P4_2/mnm	0.4433	-3.0527	0.0548	0.0603
Mo*	mp-129	Im-3m	0.3113	-1.1059	0.0139	0.0005
Mo_3Se_4*	mp-21021	R-3	0.4924	-2.4124	0.0151	0.0004
N_2	mp-1059834	Imma	1.3533	-15.9343	0.0159	0.0010
Na*	mp-10172	P6_3/mmc	0.0470	-0.0839	0.1168	0.0901
Na_2Cl	mp-1077015	Cmmm	0.0377	-0.1951	0.0901	0.0029
Na_2Cl	mp-990084	R-3m	0.0479	-0.3527	0.0618	0.0000
Na_3Cl	mp-1064484	P4/mmm	0.0297	-0.2254	0.0859	0.0249
Nb*	mp-75	Im-3m	0.3268	-1.1439	0.0000	0.0000
Nb_3Ir*	mp-1458	Pm-3n	0.4144	-2.0103	0.0000	0.0002
$NbGa_3$*	mp-1973	I4/mmm	0.3104	-1.7262	0.0000	0.0000
NbO*	mp-2692	Fm-3m	0.2933	-2.2812	0.0292	0.0023
NbO	mp-634965	P4/mmm	0.9184	-5.5839	0.0000	0.0000
NbP*	mp-9830	I4_1/amd	0.3299	-1.7803	0.0001	0.0000
NbP	mp-1220317	Cmmm	0.3238	-1.7805	0.0084	0.0004
$NbRh_2$	mp-1220414	Pm	0.3585	-1.4459	0.0440	0.0023
NbS_2	mp-774873	P6_3/mmc	0.6390	-3.1946	0.0049	0.0000
Ni*	mp-23	Fm-3m	0.0947	-0.4204	0.3799	0.2037
NiB*	mp-14019	Cmcm	0.6996	-6.1857	0.0198	0.0093
$NiHg_4$*	mp-570835	Im-3m	0.3125	-2.2179	0.0000	0.0004
NiS*	mp-1547	R3m	0.2630	-1.5895	0.1140	0.0342
Os*	mp-49	P6_3/mmc	0.2567	-1.1579	0.0538	0.0135
OsC*	mp-7142	P-6m2	0.4688	-3.0941	0.0402	0.0068
OsC	mp-1009537	Pm-3m	0.3353	-2.0943	0.0446	0.0000
OsC	mp-1009539	Fm-3m	0.4618	-3.1760	0.0828	0.0058
OsC	mp-999308	P6_3/mmc	0.4521	-3.1583	0.0525	0.0226
Pb*	mp-20483	Fm-3m	0.1943	-0.8519	0.0276	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
$PbAu_2$*	mp-19871	Fd-3m	0.3048	-2.3586	0.0000	0.0000
Pd*	mp-2	Fm-3m	0.1631	-1.2162	0.1264	0.0877
Pt*	mp-126	Fm-3m	0.2367	-1.8089	0.0927	0.0819
Rb*	mp-70	Im-3m	0.0590	-0.1180	0.1307	0.0377
Rb_2Hg_7*	mp-31474	P-3m1	0.1920	-0.9499	0.0000	0.0000
RbP*	mp-1179688	C2/m	0.1031	-0.4288	0.0138	0.0467
RbP	mp-1058499	Immm	0.1277	-0.4304	0.0099	0.0324
Re*	mp-8	P6_3/mmc	0.3162	-1.3803	0.0205	0.0030
Re_3Ge_7*	mp-29141	Cmcm	0.4553	-2.2942	0.0022	0.0002
ReC*	mp-1079494	P6_3/mmc	0.4741	-3.3056	0.0211	0.0022
ReC	mp-1009731	Pm-3m	0.3588	-2.2826	0.0149	0.0003
Rh*	mp-74	Fm-3m	0.1620	-0.7081	0.1852	0.0924
Ru*	mp-33	P6_3/mmc	0.2375	-0.9478	0.0875	0.0211
RuC*	mp-10030	P-6m2	0.4770	-2.9147	0.0418	0.0078
RuC	mp-1009787	Fm-3m	0.4752	-2.9734	0.0862	0.0067
SN	mp-1078816	P2_1/c	0.2823	-1.7903	0.0359	0.0006
SN	mp-1173453	P2_1	1.5195	-10.9244	0.0458	0.0000
SN	mp-684642	P2_1	1.3504	-13.4315	0.0333	0.0038
Sb*	mp-104	R-3m	0.4398	-1.7202	0.0425	0.0063
Sc*	mp-67	P6_3/mmc	0.2110	-0.6596	0.0000	0.0000
Sc_5Ge_3*	mp-17190	P6_3/mcm	0.3162	-1.5145	0.0000	0.0000
$ScAg_2$*	mp-1018128	I4/mmm	0.2014	-1.5516	0.0000	0.0000
$ScTc_3$	mp-867262	Fm-3m	0.3043	-1.0739	0.0080	0.0010
ScTe*	mp-10026	P6_3/mmc	0.3365	-1.5673	0.0000	0.0000
$SiAu_3$	mp-1186998	Fm-3m	0.1974	-1.4121	0.0156	0.0000
SiO_2	mp-556044	P-31c	0.7437	-6.4399	0.0117	0.0000
SiO_2	mp-638049	F4_132	0.6594	-6.0556	0.0488	0.0000
SiRh*	mp-2287	P2_1/c	0.2801	-1.6004	0.0243	0.0009
Sn*	mp-623511	I4/mmm	0.1989	-0.8803	0.0235	0.0006
Sn_7Ir_5*	mp-20466	I4mm	0.5375	-3.2800	0.0000	0.0000
Sn_7Pd_5	mp-1219006	C2	0.3765	-2.7926	0.0000	0.0004
SnPt*	mp-19856	P6_3/mmc	0.4612	-3.2535	0.0000	0.0000
Sr*	mp-139	P6_3/mmc	0.0009	-0.0171	4.9050	5.3659
Sr_2As*	mp-15697	I4/mmm	0.0787	-0.2746	0.2097	0.0267
Sr_3Sn	mp-1187136	Pm-3m	0.0501	-0.2374	0.4377	0.0368
Sr_5Sn_3*	mp-17720	I4/mcm	0.8182	-2.1022	0.1796	0.0000
$SrAl_2$*	mp-1638	Fd-3m	0.4263	-1.6649	0.0004	0.0000
SrN*	mp-1058586	Fm-3m	0.1230	-0.6070	0.0462	0.0849
SrSi*	mp-2661	Cmcm	0.8077	-2.8883	0.0564	0.0097
$SrTl_3$	mp-1206636	Pm-3m	0.2857	-1.2068	0.0201	0.0007
Ta*	mp-50	Im-3m	0.3226	-1.2794	0.0000	0.0000
Ta_4C_3	mp-1218000	R3m	0.5265	-3.5440	0.0000	0.0000
Ta_5Ge_3*	mp-17593	I4/mcm	0.3556	-2.6739	0.0000	0.0000
Ta_7Co_6*	mp-1104548	R-3m	0.4891	-3.0211	0.0018	0.0000
TaN*	mp-1497	P6/mmm	0.4241	-2.9806	0.0009	0.0000
TaN	mp-1009831	Pm-3m	0.3185	-2.1377	0.0002	0.0002
TaN	mp-1009833	F-43m	0.6451	-4.4476	0.0000	0.0006
TaP*	mp-1067587	I4_1md	0.4762	-2.7480	0.0000	0.0000
TaP	mp-1187244	P-6m2	0.4792	-2.7509	0.0000	0.0000
TaRe	mp-1217894	Cmmm	0.3311	-1.5114	0.0026	0.0003
$TaSb_2$*	mp-11697	C2/m	0.4899	-2.0334	0.0002	0.0000
Tc*	mp-113	P6_3/mmc	0.3130	-1.2061	0.0240	0.0006
TcW	mp-1217337	Cmmm	0.6035	-3.6836	0.0000	0.0002
Te_2Au*	mp-567525	C2/m	0.5036	-3.0627	0.0497	0.0327
Te_2Ir*	mp-1551	C2/m	0.7621	-4.4689	0.0230	0.0046
Te_2Pd*	mp-782	P-3m1	0.5813	-3.5060	0.0291	0.0200

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ti*	mp-46	P6_3/mmc	0.3010	-0.7536	0.0000	0.0000
Ti_3Ge*	mp-1189044	I-4	0.3243	-1.6051	0.0000	0.0000
Ti_3In*	mp-866046	P6_3/mmc	0.3128	-1.1764	0.0000	0.0000
Ti_3Pt*	mp-2339	Pm-3n	0.3632	-1.7698	0.0000	0.0000
$TiBe_{12}$*	mp-1104067	I4/mmm	0.2186	-1.9904	0.0000	0.0000
$TiMo_2$	mp-1216675	Fmmm	0.3395	-1.1740	0.0000	0.0000
TiNi*	mp-1048	P2_1/m	0.2818	-0.8685	0.0059	0.0000
TiNi	mp-1067248	Cmcm	0.2798	-0.8586	0.0060	0.0000
TiO*	mp-1071163	P-62m	0.3594	-2.4547	0.0015	0.0000
$TiSi_2$*	mp-1077503	Cmcm	0.4599	-3.0700	0.0000	0.0000
$TiSi_2$	mp-570991	Immm	0.2659	-1.2939	0.0000	0.0000
TiW	mp-1216621	Cmmm	0.6568	-3.6207	0.0000	0.0000
Tl*	mp-82	P6_3/mmc	0.1905	-1.0463	0.0000	0.0000
Tl_3Ga	mp-1187539	Fm-3m	0.2090	-1.0914	0.0000	0.0000
Tl_4Sn	mp-1216560	R-3m	0.2039	-1.3837	0.0010	0.0008
TlC*	mp-567465	Fm-3m	0.3776	-2.4265	0.0000	0.0010
TlC	mp-1058887	P4/mmm	0.4837	-3.1370	0.0000	0.0003
TlN*	mp-1017982	P6_3mc	0.6066	-3.6824	0.0023	0.0000
TlN	mp-1002222	Fm-3m	0.3446	-2.1516	0.0029	0.0000
TlN	mp-1007778	F-43m	0.6105	-3.6815	0.0017	0.0000
V*	mp-146	Im-3m	0.3178	-0.8777	0.0000	0.0000
VCo_3	mp-1095057	P6_3/mmc	0.3730	-2.5141	0.0012	0.0023
VN	mp-1017532	P6_3/mmc	0.4293	-2.6954	0.0000	0.0000
VNi_2*	mp-11531	Immm	0.4769	-1.4029	0.0918	0.0008
W*	mp-91	Im-3m	0.4372	-2.5600	0.0000	0.0001
WC*	mp-13136	Fm-3m	0.5508	-4.2240	0.0012	0.0009
WC	mp-1008630	Pm-3m	0.4127	-2.9246	0.0000	0.0010
WC	mp-1008635	F-43m	0.8710	-6.2790	0.0000	0.0000
WO_2*	mp-1524394	P2_1/c	0.6713	-5.7408	0.0039	0.0000
Y*	mp-112	P6_3/mmc	0.2374	-0.8682	0.0000	0.0000
Y_3Co	mp-1189329	P2_1	0.4718	-2.6045	0.0000	0.0000
YAg_3	mp-1187811	Fm-3m	0.1780	-1.2749	0.0000	0.0000
Zn*	mp-79	P6_3/mmc	0.2296	-1.8892	0.0000	0.0000
Zn	mp-2647117	Im-3m	0.1819	-1.3641	0.0000	0.0000
ZnAg*	mp-1912	Pm-3m	0.2035	-1.8265	0.0000	0.0000
$ZnPb_3$	mp-1187949	Fm-3m	0.2392	-1.3227	0.0004	0.0004
Zr*	mp-131	P6_3/mmc	0.3223	-1.3170	0.0000	0.0000
Zr_2Au*	mp-2348	I4/mmm	0.3095	-2.0203	0.0000	0.0000
Zr_3Be	mp-1188016	I4/mmm	0.3655	-1.7390	0.0000	0.0000
Zr_3Ge	mp-1188039	P6_3/mmc	0.3460	-1.7196	0.0000	0.0000
Zr_5Pb_3*	mp-681992	P6_3/mcm	0.3301	-1.9739	0.0000	0.0000
ZrBr*	mp-1065846	C2/m	0.3528	-2.0185	0.0000	0.0001
$ZrGa_3$*	mp-30685	I4/mmm	0.2933	-1.8532	0.0000	0.0000
$ZrMo_2$*	mp-2049	Fd-3m	0.4671	-1.6863	0.0002	0.0000
ZrRh*	mp-2808	Pm-3m	0.3062	-1.4732	0.0094	0.0024
ZrTi	mp-1215236	P4/mmm	0.3777	-1.6592	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.2 Ionic

Table S2: Ionic compounds (non-metallic, ICOBI < 0.5052, observed range 0.0444 to 0.4976) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
AgO*	mp-1079720	P2_1/c	0.2604	-2.0408	0.0297	0.0024
B_6Pb	mp-1096825	Pm-3m	0.2353	-1.9376	0.0000	0.0000
$BaCl_2$*	mp-568662	Fm-3m	0.0749	-0.5008	0.2998	0.0066
$BaCl_2$*	mp-23199	Pnma	0.0637	-0.4308	0.3027	0.0019
BaF_3*	mp-1080821	P2_1/m	0.2279	-0.7883	0.3169	0.0437
BaH_2*	mp-23715	Pnma	0.0615	-0.2536	0.1495	0.0000
BaO*	mp-1008500	P6_3/mmc	0.0788	-0.6380	0.1468	0.0016
BaO	mp-1950989	P6_3/mmc	0.1070	-0.4367	0.2225	0.0000
Be_3N_2*	mp-6977	P6_3/mmc	0.4361	-4.2580	0.0000	0.0000
Be_3N_2*	mp-18337	Ia-3	0.4759	-4.7015	0.0000	0.0000
Be_3N_2	mp-1017550	P-3m1	0.3952	-3.8753	0.0000	0.0000
$BeCl_2$	mp-1172909	I-43m	0.3697	-0.9454	0.8095	0.1761
Ca_3N_2*	mp-1047	R-3c	0.1781	-0.6015	0.1048	0.0000
Ca_3N_2*	mp-844	Ia-3	0.1820	-0.6211	0.0837	0.0000
$CaCl_2$*	mp-23214	Pnnm	0.1054	-0.7540	0.2048	0.0008
$CaCl_2$	mp-1120739	Fm-3m	0.0814	-0.5708	0.3236	0.0068
CaF_2*	mp-10464	Pnma	0.0565	-0.5566	0.2424	0.0000
CaF_2	mp-554355	P4/mmm	0.0591	-0.5440	0.1286	0.0000
CaO*	mp-1079707	Cmce	0.1008	-0.5793	0.1805	0.0000
CaO*	mp-2605	Fm-3m	0.1143	-0.7028	0.0897	0.0000
CdI_2*	mp-567259	P-3m1	0.4859	-2.4996	0.0508	0.0000
Cs_2S*	mp-1077814	P6_3/mmc	0.0804	-1.0011	0.0682	0.0000
Cs_2Se*	mp-1011696	Pnma	0.0811	-0.6189	0.0861	0.0000
CsCl*	mp-573697	Fm-3m	0.0595	-0.3453	0.0490	0.0000
CsF*	mp-8455	Pm-3m	0.1224	-1.5102	0.0523	0.0008
CsH*	mp-632319	Pm-3m	0.0935	-1.2305	0.0678	0.0002
GeS	mp-12910	Cmcm	0.4348	-2.2462	0.0516	0.0000
H_2O*	mp-2739191	C222_1	0.4372	-4.6426	0.0087	0.0000
HI*	mp-697025	C2/c	0.4915	-2.7099	0.0522	0.0000
HgF_2*	mp-8177	Fm-3m	0.2320	-1.8744	0.1660	0.0000
I*	mp-23153	Cmce	0.0444	-0.1504	0.4981	0.0900
InBr*	mp-22870	Cmcm	0.3010	-1.3292	0.0929	0.0024
InCl*	mp-571636	Cmc2_1	0.1698	-0.8368	0.0921	0.0006
K_2O*	mp-971	Fm-3m	0.1009	-0.8814	0.0059	0.0000
K_2S*	mp-559278	P6_3/mmc	0.1040	-0.4891	0.0071	0.0006
K_2S*	mp-1022	Fm-3m	0.1212	-0.4744	0.0000	0.0000
KCl*	mp-23289	Pm-3m	0.0510	-0.4078	0.1374	0.0000
KF_3*	mp-1544557	P2_12_12_1	0.3162	-0.9256	0.4749	0.3545
KF_3	mp-2749823	P2_1	0.3119	-0.9244	0.4733	0.3514
KI*	mp-22898	Fm-3m	0.0893	-0.4549	0.0615	0.0000
Li_2O*	mp-1960	Fm-3m	0.2393	-2.0320	0.0000	0.0000
Li_2O	mp-755894	Pnma	0.2225	-2.0369	0.0000	0.0000
Li_2S*	mp-1125	Pnma	0.2847	-2.0194	0.0000	0.0000
Li_2S	mp-557142	P6_3/mmc	0.3327	-2.1900	0.0000	0.0000
Li_2Se*	mp-2286	Fm-3m	0.3399	-2.1702	0.0000	0.0000
Li_3N*	mp-2341	P6_3/mmc	0.2587	-1.8972	0.0000	0.0000
LiBr	–	–	0.2790	-2.0381	0.0000	0.0000
LiCl*	mp-22905	Fm-3m	0.2558	-2.0554	0.0000	0.0000
LiCl	mp-1185319	P6_3mc	0.3879	-2.9688	0.0000	0.0000
LiF*	mp-1138	Fm-3m	0.1692	-1.9334	0.0001	0.0000
LiF	mp-1009009	Pm-3m	0.1237	-1.3411	0.0249	0.0000
$MgCl_2$*	mp-570259	P-3m1	0.1312	-0.9334	0.2905	0.0028
$MgCl_2$*	mp-23210	R-3m	0.1313	-0.9344	0.2883	0.0009

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
$MgCl_2$	mp-569626	Ama2	0.1475	-1.0096	0.1898	0.0001
MgF_2*	mp-1072956	Pnnm	0.0901	-1.0192	0.1330	0.0000
MgF_2	mp-1062842	Amm2	0.0808	-0.8465	0.1522	0.0000
MgF_2	mp-560236	C2/m	0.1007	-1.0813	0.0878	0.0000
MgH_2*	mp-23711	Pbcn	0.1156	-0.5595	0.0617	0.0000
MgO*	mp-549706	P6_3mc	0.1572	-1.2025	0.0812	0.0000
MgO	mp-1009127	Pm-3m	0.0813	-0.6164	0.2735	0.0000
MgS*	mp-1315	Fm-3m	0.1236	-0.7426	0.2438	0.0000
MgS	mp-13032	F-43m	0.2006	-1.1035	0.1624	0.0000
Na_2O*	mp-2352	Fm-3m	0.1191	-0.6209	0.0000	0.0000
Na_2O	mp-1975297	C2/m	0.1297	-0.6320	0.0003	0.0000
NaCl	–	–	0.0859	-0.5953	0.1111	0.0000
NaS*	mp-409	P-62m	0.4976	-2.6241	0.1126	0.0000
NaS	mp-1180106	C2/m	0.2104	-0.9839	0.0000	0.0000
PbSe*	mp-22009	Fmm2	0.4169	-1.6445	0.0576	0.0000
PbSe*	mp-2201	Fm-3m	0.3457	-1.4608	0.0807	0.0039
Rb_2O*	mp-1394	Fm-3m	0.0867	-1.2998	0.0368	0.0000
Rb_2O	mp-2023482	P-3m1	0.1302	-0.4067	0.0636	0.0000
RbCl*	mp-23299	Pm-3m	0.0478	-0.6732	0.1019	0.0000
RbF*	mp-2064	Pm-3m	0.0879	-1.3700	0.0447	0.0036
RbF	mp-11718	Fm-3m	0.0460	-0.4536	0.0398	0.0000
RbI*	mp-22903	Fm-3m	0.0840	-0.4162	0.0534	0.0013
RbS	mp-1179684	C2/m	0.1226	-0.5861	0.0850	0.0092
Sc_2O_3	mp-755313	R-3c	0.4210	-2.7698	0.0026	0.0000
SiO_2	mp-10064	Fm-3m	0.3974	-3.7939	0.0240	0.0000
$SnBr_2$	mp-978985	P4_2/mnm	0.3595	-1.6332	0.1289	0.0000
SrF_2*	mp-1102240	Pnma	0.0589	-0.5401	0.2090	0.0000
SrF_2	mp-1102911	Pnma	0.0505	-0.4858	0.2166	0.0000
SrS*	mp-10627	Pm-3m	0.0762	-0.4641	0.2376	0.0000
Te*	mp-19	P3_121	0.3172	-1.1233	0.1197	0.0127
Te_2Ru*	mp-267	Pnnm	0.4723	-2.2556	0.0486	0.0017
TlCl*	mp-571079	Cmcm	0.1717	-0.8409	0.1079	0.0000
Y_2O_3*	mp-558573	C2/m	0.4214	-2.9843	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.3 Covalent

Table S3: Covalent compounds (non-metallic, ICOBI $\geq$ 0.5052, observed range 0.5087 to 1.9179) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
As_2S_3*	mp-1189572	P-1	0.9136	-4.7416	0.0538	0.0000
AsF_3*	mp-28027	Pna2_1	0.6951	-6.3410	0.1114	0.0000
B*	mp-160	R-3m	0.5203	-4.7960	0.0000	0.0000
BI_3*	mp-23189	P6_3/m	1.2727	-6.8061	0.0232	0.0000
Be_3P_2*	mp-567841	Ia-3	0.5637	-3.8282	0.0000	0.0000
Be_3P_2	mp-1084771	Pn-3m	0.5501	-3.7626	0.0000	0.0002
$BeCl_2$*	mp-23267	Ibam	0.7165	-5.6352	0.0021	0.0000
$BeCl_2$	mp-1190080	I-43m	0.7528	-5.9613	0.0026	0.0000
BeF_2*	mp-15951	P3_121	0.5269	-6.8585	0.0035	0.0000
BeO*	mp-7599	P4_2/mnm	0.5148	-5.7846	0.0000	0.0000
BeO	mp-1778	F-43m	0.5232	-5.8507	0.0000	0.0000
Bi_2O_3	mp-714859	P1	1.1110	-10.1326	0.0646	0.0001

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Br*	mp-23154	Cmce	0.7827	-3.5362	0.4257	0.0097
C	mp-1080826	C2/m	0.9446	-9.5768	0.0000	0.0000
C	mp-1190171	Pnma	0.9445	-9.5798	0.0000	0.0000
C*	mp-66	Fd-3m	0.9497	-9.6602	0.0000	0.0000
C*	mp-47	P6_3/mmc	0.9482	-9.7832	0.0000	0.0000
C	–	–	1.2173	-11.6891	0.0000	0.0000
C_3N_4*	mp-571653	P-43m	0.9531	-10.4956	0.0150	0.0000
C_3N_4	mp-1182484	P6_3/m	1.9179	-15.3586	0.0000	0.0000
C_3N_4	mp-2852	I-43d	0.9575	-10.9099	0.0106	0.0000
C_3N_4	mp-563	Fd-3m	0.7047	-7.1477	0.0086	0.0000
Cl_2*	mp-22848	Cmce	0.9254	-5.5728	0.4555	0.0003
ClF_3*	mp-556767	Pnma	0.6390	-5.2173	0.1347	0.0000
F_2*	mp-561203	C2/c	0.9731	-6.9536	0.4569	0.0000
Ga_2Se_3	mp-1224809	Imm2	0.8324	-4.7105	0.0085	0.0000
$GeSe_2$*	mp-10074	I-42d	0.9011	-4.9682	0.0171	0.0000
H_2*	mp-730101	P2_12_12_1	0.9954	-5.1222	0.0000	0.0000
H_3N*	mp-643432	P2_12_12_1	0.8816	-8.0639	0.0000	0.0000
HCl	mp-684609	Imm2	0.9480	-7.3985	0.0478	0.0001
IF_7*	mp-27988	Aea2	0.5305	-5.5662	0.0568	0.0000
$IrCl_3$	mp-729507	C2	0.7112	-4.9995	0.0957	0.0000
Mn_2O_7*	mp-28338	P2_1/c	1.3609	-6.7339	0.0257	0.0000
MnN	mp-1009130	F-43m	0.6682	-3.9792	0.0247	0.0062
$MoSe_2$*	mp-1634	P6_3/mmc	0.6411	-2.8467	0.0035	0.0000
Nb_2O_5*	mp-1595	C2/m	0.7548	-4.9296	0.0075	0.0000
O_2*	mp-1524462	C2/m	1.4816	-17.7956	0.1271	0.2186
P*	mp-568348	P2/c	0.9202	-5.0760	0.0209	0.0000
PS*	mp-612	C2/c	0.9840	-5.8992	0.0301	0.0000
$PtCl_2$	mp-1219748	C2/m	0.7183	-5.3013	0.0756	0.0000
S*	mp-77	Fddd	0.9692	-5.9246	0.0880	0.0000
SN*	mp-236	P2_1/c	1.1480	-8.7413	0.0440	0.0000
Se*	mp-14	P3_121	0.9053	-4.1496	0.0828	0.0000
Si	–	–	0.9243	-4.5418	0.0000	0.0000
SiO_2*	mp-546794	I-42d	0.7466	-7.9383	0.0071	0.0000
SiO_2*	mp-9258	Pa-3	0.5130	-5.5969	0.0264	0.0000
SiS_2*	mp-1095294	P2_1/c	0.9089	-5.7585	0.0044	0.0000
TeO_2*	mp-8377	P2_12_12_1	0.6215	-4.7653	0.0772	0.0000
TiO_2*	mp-1439	Pbcn	0.5838	-3.5796	0.0077	0.0000
TiO_2*	mp-9173	Pnma	0.5897	-3.5580	0.0123	0.0000
TiO_2*	mp-430	P2_1/c	0.5087	-3.0819	0.0145	0.0000
ZrO_2	mp-775909	P4_2/mnm	0.5720	-4.1446	0.0017	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.4 Mixed (Zintl-type)

Table S4: Mixed-bonding (Zintl-type) compounds (a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide N_3^- ion ionically bound to its cation; observed range 0.8522 to 2.7322) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ba_5P_4*	mp-17566	Pnma	0.9040	-3.4927	0.2104	0.0000
BaS_3*	mp-556296	P2_12_12	0.9201	-5.2850	0.1084	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Bi_2O_3	mp-674644	P1	0.9332	-7.5456	0.0131	0.0000
CdO_2^*	mp-2310	Pa-3	0.9738	-7.1680	0.0540	0.0000
$Cs_2As_3^*$	mp-15557	Fmmm	1.0431	-4.4248	0.0386	0.0000
CsN_3^*	mp-510557	I4/mcm	1.8888	-17.7624	0.0000	0.0000
CsN_3	mp-1225892	P4/mmm	1.8983	-17.5725	0.0000	0.0000
CsP_7^*	mp-1201790	Pbcm	0.9343	-4.9186	0.0062	0.0000
CsP_7	mp-1213206	Pca2_1	0.9336	-4.9318	0.0064	0.0000
KN_3^*	mp-827	I4/mcm	1.8889	-18.4973	0.0000	0.0000
LiP_5^*	mp-2412	Pna2_1	0.9037	-4.9376	0.0026	0.0000
LiP_5	mp-32760	Pna2_1	0.8771	-4.6811	0.0064	0.0000
MgP_4^*	mp-384	P2_1/c	0.9196	-5.0216	0.0273	0.0000
NaN_3^*	mp-1066400	R3m	1.8853	-18.8113	0.0000	0.0000
NaN_3^*	mp-22003	R-3m	1.8872	-19.1626	0.0002	0.0000
NaN_3	mp-1065265	C2/m	2.7322	-23.2128	0.0006	0.0215
NaS^*	mp-2400	P6_3/mmc	0.9510	-4.7236	0.1279	0.0000
PI_2^*	mp-29443	P-1	0.9586	-4.8005	0.1081	0.0000
PbN_6^*	mp-667338	Pnma	1.8687	-19.2227	0.0007	0.0000
RbN_3^*	mp-581833	P4/mmm	1.8969	-18.3403	0.0000	0.0000
RbN_3	mp-1219567	Amm2	1.3756	-9.8572	0.0216	0.0000
RbS^*	mp-558071	Immm	0.9920	-5.0554	0.1964	0.0019
Sr_3As_4	mp-678007	P1	0.8522	-3.0556	0.1161	0.0000
$Sr_3P_4^*$	mp-14288	Fdd2	0.8910	-4.1353	0.1224	0.0000
Sr_3P_4	mp-682953	P1	0.8889	-4.0705	0.1250	0.0000
SrO_2^*	mp-2697	I4/mmm	0.9835	-7.0573	0.1841	0.0000
SrO_2^*	mp-726705	Pnma	0.9821	-6.8700	0.2389	0.0000
SrO_2	mp-2763179	Pnma	0.9805	-6.9738	0.2327	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.5 Unclassified

Table S5: Unclassified compounds (no ICOBI data available) — same columns, ICOBI/antibonding shown as "–" where missing.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

B VASP Calculation Parameters

Fixed INCAR parameters, identical across the whole dataset (static, LOBSTER-compatible run) — only `NBANDS` (appendix D) and the k-point grid vary by compound.

Parameter	Value	Description
`PREC`	Accurate	Overall precision (charge grid, projectors)
`ENCUT`	600.0	Plane-wave cutoff energy (eV)
`EDIFF`	1e-06	Electronic convergence criterion (eV)
`IBRION`	-1	Ionic dynamics type (-1 = static, no relaxation)
`NSW`	0	Number of ionic steps (0 = static run on the MP structure)
`ISMEAR`	0	Smearing type (0 = Gaussian)
`SIGMA`	0.05	Smearing width (eV)
`LASPH`	True	Non-spherical corrections to the PAW potential
`ISYM`	-1	Symmetry (-1 = disabled, required by LOBSTER)
`LWAVE`	True	Write WAVECAR (required by LOBSTER)
`LCHARG`	True	Write CHGCAR
`NPAR`	4	Band-level parallelization
`KPAR`	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen's `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table S6: PAW-PBE potentials used, one per element present in the dataset (pymatgen's MPRelaxSet library, with one documented exception: W substituted for W_pv/W_sv, see §7).

Element	PAW-PBE potential
Ag	`Ag (02Apr2005)`
Al	`Al (04Jan2001)`
As	`As (22Sep2009)`
Au	`Au (04Oct2007)`
B	`B (06Sep2000)`
Ba	`Ba_sv (06Sep2000)`
Be	`Be_sv (06Sep2000)`
Bi	`Bi (08Apr2002)`
Br	`Br (06Sep2000)`
C	`C (08Apr2002)`
Ca	`Ca_sv (06Sep2000)`
Cd	`Cd (06Sep2000)`
Cl	`Cl (06Sep2000)`
Co	`Co (02Aug2007)`
Cr	`Cr_pv (02Aug2007)`
Cs	`Cs_sv (25Jan2019)`
Cu	`Cu_pv (06Sep2000)`
F	`F (08Apr2002)`
Fe	`Fe_pv (02Aug2007)`
Ga	`Ga_d (06Jul2010)`
Ge	`Ge_d (03Jul2007)`
H	`H (15Jun2001)`

Element	PAW-PBE potential
Hf	`Hf_pv (06Sep2000)`
Hg	`Hg (06Sep2000)`
I	`I (08Apr2002)`
In	`In_d (06Sep2000)`
Ir	`Ir (06Sep2000)`
K	`K_sv (06Sep2000)`
Li	`Li_sv (10Sep2004)`
Mg	`Mg_pv (13Apr2007)`
Mn	`Mn_pv (02Aug2007)`
Mo	`Mo_pv (04Feb2005)`
N	`N (08Apr2002)`
Na	`Na_pv (19Sep2006)`
Nb	`Nb_pv (08Apr2002)`
Ni	`Ni_pv (06Sep2000)`
O	`O (08Apr2002)`
Os	`Os_pv (20Jan2003)`
P	`P (06Sep2000)`
Pb	`Pb_d (06Sep2000)`
Pd	`Pd (04Jan2005)`
Pt	`Pt (04Feb2005)`
Rb	`Rb_sv (06Sep2000)`
Re	`Re_pv (06Sep2000)`
Rh	`Rh_pv (25Jan2005)`
Ru	`Ru_pv (28Jan2005)`
S	`S (06Sep2000)`
Sb	`Sb (06Sep2000)`
Sc	`Sc_sv (07Sep2000)`
Se	`Se (06Sep2000)`
Si	`Si (05Jan2001)`
Sn	`Sn_d (06Sep2000)`
Sr	`Sr_sv (07Sep2000)`
Ta	`Ta_pv (07Sep2000)`
Tc	`Tc_pv (04Feb2005)`
Te	`Te (08Apr2002)`
Ti	`Ti_pv (07Sep2000)`
Tl	`Tl_d (06Sep2000)`
V	`V_pv (07Sep2000)`
W	`W (08Apr2002)`
Y	`Y_sv (25May2007)`
Zn	`Zn (06Sep2000)`
Zr	`Zr_sv (04Jan2005)`

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ-centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table S7: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As_2S_3	mp-1189572	5 3 2 (G)	80
AsF_3	mp-28027	5 4 4 (G)	64
BI_3	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
$BaAu_2$	mp-30363	6 6 7 (G)	23
$BaCl_2$	mp-568662	6 6 6 (G)	13
BaH_2	mp-23715	6 4 3 (G)	28
Be_2Cu	mp-2031	7 7 7 (G)	38
Be_2Nb_3	mp-11272	8 4 4 (M)	74
Be_2V	mp-11281	7 7 4 (G)	76
$CaAl_4$	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs_2As_3	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge_2Os	mp-10032	9 6 4 (G)	54
$HfTc_2$	mp-1095669	5 5 3 (G)	108
Hg_4Pt	mp-936	5 5 5 (G)	45
HgF_2	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN_3	mp-827	5 5 5 (G)	34
Li_2Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg_2Ni	mp-2137	5 5 2 (G)	102
MgP_4	mp-384	5 5 3 (G)	40
$MnAl_{12}$	mp-1104165	4 4 4 (M)	57
$MoSe_2$	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb_3Ir	mp-1458	5 5 5 (G)	72
$NbGa_3$	mp-1973	8 8 6 (M)	36
PI_2	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc_5Ge_3	mp-17190	3 3 5 (G)	154
$ScAg_2$	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO_2	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta_5Ge_3	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
$TaSb_2$	mp-11697	8 5 4 (G)	34
Te_2Pd	mp-782	7 7 5 (G)	17
Te_2Ru	mp-267	7 5 4 (G)	34
TeO_2	mp-8377	6 5 3 (G)	48
Ti_3In	mp-866046	5 5 6 (G)	72
Ti_3Pt	mp-2339	5 5 5 (G)	72
$TiBe_{12}$	mp-1104067	7 5 5 (G)	69

Table S7 (continued)

Formula	mp-id	k-point grid	NBANDS
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
$TiSi_2$	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi_2	mp-11531	12 8 7 (G)	27
WO_2	mp-1524394	6 5 5 (G)	68
Zr_2Au	mp-2348	9 9 4 (G)	29
Zr_5Pb_3	mp-681992	3 3 5 (G)	154
$ZrMo_2$	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI_2	mp-567259	7 7 4 (G)	17
$MgCl_2$	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li_3Hg	mp-1646	7 7 7 (G)	24
Ta_7Co_6	mp-1104548	6 6 3 (G)	117
$CaGa_4$	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH_2	mp-23711	6 5 5 (G)	24
Sn_7Ir_5	mp-20466	4 4 4 (M)	108
NaN_3	mp-1066400	7 7 7 (G)	16
ClF_3	mp-556767	6 4 3 (G)	64
SiS_2	mp-1095294	5 4 3 (G)	48
Sr_5Sn_3	mp-17720	3 3 3 (G)	104
Te_2Ir	mp-1551	7 5 2 (G)	51
$Li_{13}Sn_5$	mp-30769	6 6 1 (G)	110
H_3N	mp-643432	8 5 5 (G)	28
Cs_2Se	mp-1011696	5 3 2 (G)	56
In_3Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb_2Hg_7	mp-31474	4 4 4 (G)	73
$PbAu_2$	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO_2	mp-546794	6 6 6 (M)	24
IF_7	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al_3Os_2	mp-16521	9 9 3 (G)	30
Nb_2O_5	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
$MgPd_3$	mp-7753	7 7 7 (G)	31
$AsPd_2$	mp-1080831	8 4 4 (G)	66
$GeSe_2$	mp-10074	5 5 5 (G)	34
Re_3Ge_7	mp-29141	9 6 1 (G)	180
Cu_3Ge	mp-19724	6 6 5 (G)	72
Ti_3Ge	mp-1189044	4 4 4 (M)	144
$ZrGa_3$	mp-30685	7 7 5 (G)	37
$SrAl_2$	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
$Al_{12}W$	mp-11227	4 4 4 (M)	57

Table S7 (continued)

Formula	mp-id	k-point grid	NBANDS
CsH	mp-632319	7 7 7 (G)	6
$CaGa_2$	mp-992	7 7 6 (G)	23
$NiHg_4$	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
$BaSn_5$	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te_2Au	mp-567525	7 7 5 (G)	17
Be_3N_2	mp-6977	10 10 3 (G)	46
Y_2O_3	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
$FeTe_2$	mp-1102002	4 4 4 (M)	68
H_2O	mp-2739191	7 7 4 (G)	24
CdO_2	mp-2310	5 5 5 (G)	68
Mo_3Se_4	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
$Be_{12}Pd$	mp-12666	6 6 6 (M)	69
Sr_2As	mp-15697	6 6 3 (G)	28
$AsRh_2$	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K_2S	mp-559278	5 5 4 (G)	28
Ge_4Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li_3Ag	mp-976408	7 7 7 (G)	24
Y_3Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr_3As_4	mp-678007	3 3 4 (G)	62
Ga_2Se_3	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
$BeCl_2$	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
$SrTl_3$	mp-1206636	5 5 5 (G)	32
$IrCl_3$	mp-729507	3 3 6 (G)	84
Ta_4C_3	mp-1218000	6 6 6 (M)	48
Sc_2O_3	mp-755313	6 5 6 (G)	64
$SnBr_2$	mp-978985	4 4 6 (M)	34
Sn_7Pd_5	mp-1219006	5 5 3 (G)	108
Tl_4Sn	mp-1216560	5 5 5 (G)	45
$LiTl_3$	mp-1185253	6 6 6 (M)	32
$PtCl_2$	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K_5As_4	mp-684799	4 4 3 (G)	41
KSn_3	mp-1185151	4 4 5 (G)	64
NbS_2	mp-774873	10 6 2 (M)	34
YAg_3	mp-1187811	6 6 6 (G)	37
B_3Ir_2	mp-1228647	9 5 4 (G)	60
$TiMo_2$	mp-1216675	5 6 14 (G)	27
Cr_2C	mp-1226378	10 10 7 (G)	22
ZrO_2	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN_2	mp-1097011	5 5 5 (G)	24

Table S7 (continued)

Formula	mp-id	k-point grid	NBANDS
$CoSe_2$	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo_3	mp-1095057	7 6 6 (G)	72
Tl_3Ga	mp-1187539	5 5 5 (G)	36
$CaGe_2$	mp-643542	7 7 6 (G)	23
$NbRh_2$	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF_2	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na_2Cl	mp-990084	6 6 6 (M)	12
$ZnPb_3$	mp-1187949	5 5 5 (G)	36
Sr_3Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al_4Si	mp-1228120	6 6 6 (M)	20
B_6Pb	mp-1096825	6 6 6 (M)	33
$ScTc_3$	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd_5	mp-1228665	6 6 6 (M)	49
Zr_3Be	mp-1188016	6 6 6 (M)	35
Na_3Cl	mp-1064484	8 8 3 (G)	16
$SiAu_3$	mp-1186998	7 7 7 (G)	31
$InSe_2$	mp-1232036	7 7 3 (G)	34
$TiSi_2$	mp-570991	8 8 8 (M)	17
$BeAu_3$	mp-983539	7 7 7 (G)	32
Zr_3Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
$Mg_{15}B$	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E Elements and Allotropes

Every single-element structure with no matched polymorph in the dataset: the elemental references (`ELEMENT_REFERENCE`) used to decompose multi-element compounds. An experimental compound's formula is marked with a superscript star (black if it is the stable phase, red if it is a metastable polymorph/allotrope); unmarked formulas are theoretical. Auto-generated by `report/gen_appendix_extension.py`.

Table S8: Elements and allotropes with no matched polymorph in the `extension` campaign: references used to decompose multi-element compounds. Allotropes with one or more other allotropes computed in this dataset (e.g. carbon) are grouped in Table S9 instead. Black * = stable experimental reference (thermodynamic phase); red * = metastable experimental reference; unmarked = theoretical. No ΔICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6_3/mmc	0.0000
Br*	mp-23154	Cmce	0.0000
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6_3/mmc	0.0000
Cl_2*	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F_2*	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H_2*	mp-730101	P2_12_12_1	0.0000
Hf*	mp-103	P6_3/mmc	0.0000
Hg*	mp-10861	P6/mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4/mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6_3/mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N_2	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O_2*	mp-1524462	C2/m	0.0000
Os*	mp-49	P6_3/mmc	0.0000
P*	mp-568348	P2/c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092
Re*	mp-8	P6_3/mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6_3/mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000
Sc*	mp-67	P6_3/mmc	0.0000
Se*	mp-14	P3_121	0.0011
Sn*	mp-623511	I4/mmm	0.0614
Sr*	mp-139	P6_3/mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6_3/mmc	0.0000
Te*	mp-19	P3_121	0.0000
Ti*	mp-46	P6_3/mmc	0.0152
Tl*	mp-82	P6_3/mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000
Y*	mp-112	P6_3/mmc	0.0000
Zn*	mp-79	P6_3/mmc	0.0000
Zr*	mp-131	P6_3/mmc	0.0000

F Polymorphs and Allotropes

Elements and compounds with several entries sharing a formula in the dataset, grouped here rather than scattered across the elements/compounds tables (see the table caption for detail).

Table S9: Elements (allotropes) and compounds (polymorphs) with **several entries sharing a formula** in the `extension` campaign (54 groups, 130 entries) — grouped here rather than scattered across the elements/compounds tables, since the natural comparison for a polymorph group is polymorph-to-polymorph, not the decomposition-to-elements reaction (hence no ΔICOHP column). A rule separates each group; within a group, sorted by increasing E_{hull}. Black * = stable experimental; red * = metastable experimental; unmarked = theoretical.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
AgN*	mp-1091417	Cmce	1.5201
AgN	mp-1009766	F-43m	1.5495
AgN	mp-1428312	Pm-3m	1.6519
AgN	mp-1424734	Ama2	1.9196
BaF_3*	mp-1080821	P2_1/m	0.0000
BaF_3	mp-1183303	I4/mmm	0.0612
BaO*	mp-1008500	P6_3/mmc	0.0189
BaO	mp-1950989	P6_3/mmc	0.0434
Be_3N_2*	mp-18337	Ia-3	0.0000
Be_3N_2	mp-1017550	P-3m1	0.2081
Be_3P_2*	mp-567841	Ia-3	0.0000
Be_3P_2	mp-1084771	Pn-3m	0.1567
$BeCl_2$*	mp-23267	Ibam	0.0087
$BeCl_2$	mp-1172909	I-43m	3.1688
BeF_2*	mp-15951	P3_121	0.0004
BeF_2	mp-975644	P6/mmm	2.5181
BeO	mp-1778	F-43m	0.0070
BeO*	mp-7599	P4_2/mnm	0.0153
Bi_2O_3*	mp-558953	P2_1/c	0.6000
Bi_2O_3	mp-674644	P1	0.6199
Bi_2O_3	mp-1383505	P1	0.9068
Bi_2O_3	mp-714859	P1	2.4029
C*	mp-48	P6_3/mmc	0.0031
C*	mp-66	Fd-3m	0.1123
C*	mp-47	P6_3/mmc	0.1395
C	mp-1190171	Pnma	0.2927
C	mp-1080826	C2/m	0.3009
C_3N_4*	mp-571653	P-43m	0.4981
C_3N_4	mp-2852	I-43d	0.5033
C_3N_4	mp-1182484	P6_3/m	0.5848
C_3N_4	mp-563	Fd-3m	1.5119
Ca_3N_2*	mp-844	Ia-3	0.0000
Ca_3N_2*	mp-1047	R-3c	0.0167
Ca_3N_2	mp-1013524	Pm-3m	0.5112
$CaCl_2$*	mp-23214	Pnnm	0.0011

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
$CaCl_2$	mp-1120739	Fm-3m	0.0422
CaF_2*	mp-10464	Pnma	0.0346
CaF_2	mp-554355	P4/mmm	0.2403
CaO*	mp-2605	Fm-3m	0.0000
CaO*	mp-1079707	Cmce	0.2098
CaO	mp-1181903	Fd-3m	2.7281
CsN_3*	mp-510557	I4/mcm	0.0000
CsN_3	mp-1225892	P4/mmm	0.0113
CsO_2*	mp-1441	I4/mmm	0.0349
CsO_2	mp-1096936	Cmmm	0.0940
CsP_7*	mp-1201790	Pbcm	0.0000
CsP_7	mp-1213206	Pca2_1	0.0001
K_2O*	mp-971	Fm-3m	0.0000
K_2O	mp-1223784	P4/mmm	0.2000
KF_3	mp-2749823	P2_1	0.0119
KF_3*	mp-1544557	P2_12_12_1	0.0125
KN_2*	mp-1071868	Cmc2_1	1.1630
KN_2	mp-1180757	Cmcm	1.2293
KP*	mp-1058315	R-3m	1.0080
KP	mp-1057787	Immm	1.0092
Li_2O*	mp-1960	Fm-3m	0.0000
Li_2O	mp-755894	Pnma	0.0844
Li_2S*	mp-1125	Pnma	0.0615
Li_2S	mp-557142	P6_3/mmc	0.1750
LiCl	mp-1185319	P6_3mc	0.0039
LiCl*	mp-22905	Fm-3m	0.0243
LiF*	mp-1138	Fm-3m	0.0000
LiF	mp-1009009	Pm-3m	0.2875
LiP_5*	mp-2412	Pna2_1	0.0077
LiP_5	mp-32760	Pna2_1	0.0963
$MgCl_2$*	mp-23210	R-3m	0.0000
$MgCl_2$	mp-569626	Ama2	0.1991
MgF_2*	mp-1072956	Pnnm	0.0025
MgF_2	mp-1062842	Amm2	0.2636
MgO*	mp-549706	P6_3mc	0.1180
MgO	mp-1009127	Pm-3m	0.7484
MgS*	mp-1315	Fm-3m	0.0000
MgS	mp-13032	F-43m	0.0343
Na_2O*	mp-2352	Fm-3m	0.0000
Na_2O	mp-1975297	C2/m	0.0981
NaN_3*	mp-22003	R-3m	0.0017
NaN_3	mp-1065265	C2/m	1.0162
NaS*	mp-409	P-62m	0.0023
NaS	mp-1180106	C2/m	1.0061
NbO*	mp-2692	Fm-3m	0.7888
NbO	mp-634965	P4/mmm	0.8240

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
NbP*	mp-9830	I4_1/amd	0.6021
NbP	mp-1220317	Cmmm	0.6209
OsC*	mp-7142	P-6m2	0.9161
OsC	mp-999308	P6_3/mmc	0.9278
OsC	mp-1009539	Fm-3m	1.3918
OsC	mp-1009537	Pm-3m	1.4795
Rb_2O*	mp-1394	Fm-3m	0.0032
Rb_2O	mp-2023482	P-3m1	0.0479
RbF	mp-11718	Fm-3m	0.0000
RbF*	mp-2064	Pm-3m	0.0516
RbN_3*	mp-581833	P4/mmm	0.0193
RbN_3	mp-1219567	Amm2	0.7966
RbP*	mp-1179688	C2/m	0.9879
RbP	mp-1058499	Immm	0.9882
RbS*	mp-558071	Immm	0.0019
RbS	mp-1179684	C2/m	0.9975
ReC*	mp-1079494	P6_3/mmc	0.6362
ReC	mp-1009731	Pm-3m	1.1150
RuC*	mp-10030	P-6m2	0.6494
RuC	mp-1009787	Fm-3m	0.9097
SN*	COD:7017102	P2_1/c	–
SN*	mp-236	P2_1/c	0.5087
SN	mp-684642	P2_1	0.5498
SN	mp-1173453	P2_1	0.7225
SN	mp-1078816	P2_1/c	0.8090
SiO_2*	mp-9258	Pa-3	0.5770
SiO_2	mp-10064	Fm-3m	1.2779
SiO_2	mp-556044	P-31c	1.4574
SiO_2	mp-638049	F4_132	1.7448
Sr_3P_4*	mp-14288	Fdd2	0.0000
Sr_3P_4	mp-682953	P1	0.0081
SrF_2	mp-1102911	Pnma	0.0267
SrF_2*	mp-1102240	Pnma	0.0646
SrO_2*	mp-726705	Pnma	0.0096
SrO_2	mp-2763179	Pnma	0.0110
TaN*	mp-1497	P6/mmm	0.4516
TaN	mp-1009833	F-43m	0.4917
TaN	mp-1009831	Pm-3m	0.8163
TiO_2*	mp-1439	Pbcn	0.0045
TiO_2*	mp-430	P2_1/c	0.0395
TiO_2*	mp-9173	Pnma	0.0575
TlC*	mp-567465	Fm-3m	2.1080
TlC	mp-1058887	P4/mmm	2.3757
TlN*	mp-1017982	P6_3mc	0.7334
TlN	mp-1007778	F-43m	0.7337
TlN	mp-1002222	Fm-3m	0.9712
WC*	mp-13136	Fm-3m	0.4477
WC	mp-1008635	F-43m	0.6729

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
WC	mp-1008630	Pm-3m	0.8990

G Target Compounds

Every multi-element target compound with no matched polymorph in the dataset, with ΔICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_case1_full.csv` by `report/gen_appendix_extension.py`.

Table S10: Target compounds with no matched polymorph in the `extension` campaign. Formulas with several entries in this dataset are grouped in Table S9 instead. Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ΔICOHP per atom, products − reactants convention (`reaction_analysis`); "–" where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	ΔICOHP/at. (eV)	Label
Ba_5P_4*	mp-17566	Pnma	0.0000	-0.9590	exobondic
$BaCl_2$*	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaN_2*	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaS_3*	mp-556296	P2_12_12	0.0436	-0.6233	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
Cs_2S*	mp-1077814	P6_3/mmc	0.0667	0.9554	endobondic
CsCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CsF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
K_2S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KN_3	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
Li_3N*	mp-2341	P6_3/mmc	0.0039	0.6165	endobondic
Mn_2O_7*	mp-28338	P2_1/c	0.3346	-1.4687	exobondic
MnO_2*	mp-510408	P4_2/mnm	0.0464	-0.5274	exobondic
Na_2Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
PbN_6*	mp-667338	Pnma	0.1093	-2.5257	exobondic
RbCl*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
S_2N*	COD:4031496	P4_2nm	–	-0.9767	exobondic
SrN*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrS*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic

H Endobondic Reactions

The subset of case-1 decomposition reactions ($A_aB_b \rightarrow a\,A + b\,B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** (ΔICOHP ≥ 0, products − reactants convention — note this is the sign *opposite* `reaction_icohp.py`'s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`'s docstring).

Table S11: Decomposition-into-elements (case 1) reactions labeled **endobondic** (ΔICOHP ≥ 0, products − reactants) among the 208 compounds of the `extension` campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
0.3333 $Be_3N_2 \rightarrow Be + 0.3333\ N_2$	mp-1017550	0.2081	0.3960
$BeCl_2{}^* \rightarrow Be + Cl_2$	mp-23267	0.0087	1.8204
$BeF_2{}^* \rightarrow Be + F_2$	mp-15951	0.0004	3.1514
$BeO^* \rightarrow Be + 0.5\ O_2$	mp-7599	0.0153	2.7864
$BeO \rightarrow Be + 0.5\ O_2$	mp-1778	0.0070	4.4391
0.3333 $C_3N_4{}^* \rightarrow C + 0.6667\ N_2$	mp-571653	0.4981	1.1208
0.3333 $C_3N_4 \rightarrow C + 0.6667\ N_2$	mp-2852	0.5033	1.8456
0.5 $Cs_2S^* \rightarrow Cs + 0.5\ S$	mp-1077814	0.0667	0.9554
$CsCl^* \rightarrow Cs + 0.5\ Cl_2$	mp-573697	0.0071	0.8830
$CsF^* \rightarrow Cs + 0.5\ F_2$	mp-8455	0.1088	3.8464
$CsO_2{}^* \rightarrow Cs + O_2$	mp-1441	0.0349	0.0666
$CsO_2 \rightarrow Cs + O_2$	mp-1096936	0.0940	0.7743
$KCl^* \rightarrow K + 0.5\ Cl_2$	mp-23289	0.0776	0.7616
0.5 $Li_2O^* \rightarrow Li + 0.25\ O_2$	mp-1960	0.0000	2.3366
0.5 $Li_2O \rightarrow Li + 0.25\ O_2$	mp-755894	0.0844	1.9481
0.5 $Li_2S^* \rightarrow Li + 0.5\ S$	mp-1125	0.0615	1.8505
0.5 $Li_2S \rightarrow Li + 0.5\ S$	mp-557142	0.1750	1.5052
0.3333 $Li_3N^* \rightarrow Li + 0.1667\ N_2$	mp-2341	0.0039	0.6165
$LiCl^* \rightarrow Li + 0.5\ Cl_2$	mp-22905	0.0243	2.8030
$LiCl \rightarrow Li + 0.5\ Cl_2$	mp-1185319	0.0039	2.5005
$LiF^* \rightarrow Li + 0.5\ F_2$	mp-1138	0.0000	4.4673
$LiF \rightarrow Li + 0.5\ F_2$	mp-1009009	0.2875	4.5131
$LiP_5{}^* \rightarrow Li + 5\ P$	mp-2412	0.0077	0.9043
$LiP_5 \rightarrow Li + 5\ P$	mp-32760	0.0963	0.8692
$MgCl_2{}^* \rightarrow Mg + Cl_2$	mp-23210	0.0000	0.1568
$MgF_2{}^* \rightarrow Mg + F_2$	mp-1072956	0.0025	0.0015
$MgF_2 \rightarrow Mg + F_2$	mp-1062842	0.2636	0.0229
0.5 $Rb_2O^* \rightarrow Rb + 0.25\ O_2$	mp-1394	0.0032	0.7235
$RbCl^* \rightarrow Rb + 0.5\ Cl_2$	mp-23299	0.0298	1.7924
$RbF^* \rightarrow Rb + 0.5\ F_2$	mp-2064	0.0516	3.9244
$RbF \rightarrow Rb + 0.5\ F_2$	mp-11718	0.0000	1.9497
$SiO_2{}^* \rightarrow Si + O_2$	mp-9258	0.5770	3.3332
$SiO_2 \rightarrow Si + O_2$	mp-10064	1.2779	2.1735
$SiO_2 \rightarrow Si + O_2$	mp-556044	1.4574	0.0958
$SiO_2 \rightarrow Si + O_2$	mp-638049	1.7448	0.3849

I Exobondic Reactions

The complementary **exobondic** subset (ΔICOHP < 0), same convention and provenance as Appendix H.

Table S12: Decomposition-into-elements (case 1) reactions labeled **exobondic** (ΔICOHP < 0, products − reactants) among the 208 compounds of the `extension` campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
$AgN^* \rightarrow Ag + 0.5\ N_2$	mp-1091417	1.5201	-7.5875
$AgN \rightarrow Ag + 0.5\ N_2$	mp-1009766	1.5495	-6.4957

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
$AgN \rightarrow Ag + 0.5\ N_2$	mp-1424734	1.9196	-8.0907
$AgN \rightarrow Ag + 0.5\ N_2$	mp-1428312	1.6519	-5.3559
$0.2\ Ba_5P_4{}^* \rightarrow Ba + 0.8\ P$	mp-17566	0.0000	-0.9590
$BaCl_2{}^* \rightarrow Ba + Cl_2$	mp-23199	0.0068	-0.2244
$BaF_3{}^* \rightarrow Ba + 1.5\ F_2$	mp-1080821	0.0000	-0.8680
$BaF_3 \rightarrow Ba + 1.5\ F_2$	mp-1183303	0.0612	-0.2493
$BaN_2{}^* \rightarrow Ba + N_2$	mp-1102371	2.0568	-8.6248
$BaO^* \rightarrow Ba + 0.5\ O_2$	mp-1008500	0.0189	-0.3221
$BaO \rightarrow Ba + 0.5\ O_2$	mp-1950989	0.0434	-1.6129
$BaS_3{}^* \rightarrow Ba + 3\ S$	mp-556296	0.0436	-0.6233
$0.3333\ Be_3N_2{}^* \rightarrow Be + 0.3333\ N_2$	mp-18337	0.0000	-0.0746
$0.3333\ Be_3P_2{}^* \rightarrow Be + 0.6667\ P$	mp-567841	0.0000	-0.2111
$0.3333\ Be_3P_2 \rightarrow Be + 0.6667\ P$	mp-1084771	0.1567	-0.1795
$BeCl_2 \rightarrow Be + Cl_2$	mp-1172909	3.1688	-5.7807
$BeF_2 \rightarrow Be + F_2$	mp-975644	2.5181	-0.6816
$0.3333\ C_3N_4 \rightarrow C + 0.6667\ N_2$	mp-1182484	0.5848	-3.6399
$0.3333\ C_3N_4 \rightarrow C + 0.6667\ N_2$	mp-563	1.5119	-0.4498
$0.3333\ Ca_3N_2{}^* \rightarrow Ca + 0.3333\ N_2$	mp-1047	0.0167	-4.4881
$0.3333\ Ca_3N_2{}^* \rightarrow Ca + 0.3333\ N_2$	mp-844	0.0000	-4.4374
$0.3333\ Ca_3N_2 \rightarrow Ca + 0.3333\ N_2$	mp-1013524	0.5112	-4.4898
$CaCl_2{}^* \rightarrow Ca + Cl_2$	mp-23214	0.0011	-0.3199
$CaCl_2 \rightarrow Ca + Cl_2$	mp-1120739	0.0422	-0.2208
$CaF_2{}^* \rightarrow Ca + F_2$	mp-10464	0.0346	-0.2227
$CaF_2 \rightarrow Ca + F_2$	mp-554355	0.2403	-0.1814
$CaN^* \rightarrow Ca + 0.5\ N_2$	mp-1058549	0.3982	-4.9183
$CaO^* \rightarrow Ca + 0.5\ O_2$	mp-1079707	0.2098	-2.2127
$CaO^* \rightarrow Ca + 0.5\ O_2$	mp-2605	0.0000	-0.9525
$CaO \rightarrow Ca + 0.5\ O_2$	mp-1181903	2.7281	-2.9971
$CsN_3{}^* \rightarrow Cs + 1.5\ N_2$	mp-510557	0.0000	-2.2903
$CsN_3 \rightarrow Cs + 1.5\ N_2$	mp-1225892	0.0113	-1.6664
$CsP_7{}^* \rightarrow Cs + 7\ P$	mp-1201790	0.0000	-0.2978
$CsP_7 \rightarrow Cs + 7\ P$	mp-1213206	0.0001	-0.2667
$0.5\ K_2O^* \rightarrow K + 0.25\ O_2$	mp-971	0.0000	-0.4357
$0.5\ K_2O \rightarrow K + 0.25\ O_2$	mp-1223784	0.2000	-1.5638
$0.5\ K_2S^* \rightarrow K + 0.5\ S$	mp-1022	0.0000	-0.3359
$KF_3{}^* \rightarrow K + 1.5\ F_2$	mp-1544557	0.0125	-1.1511
$KF_3 \rightarrow K + 1.5\ F_2$	mp-2749823	0.0119	-1.1505
$KN_2{}^* \rightarrow K + N_2$	mp-1071868	1.1630	-3.4244
$KN_2 \rightarrow K + N_2$	mp-1180757	1.2293	-4.2865
$KN_3 \rightarrow K + 1.5\ N_2$	mp-636056	1.0370	-5.9991
$KP^* \rightarrow K + P$	mp-1058315	1.0080	-2.1600
$KP \rightarrow K + P$	mp-1057787	1.0092	-2.1366
$MgCl_2 \rightarrow Mg + Cl_2$	mp-569626	0.1991	-0.0862
$MgO^* \rightarrow Mg + 0.5\ O_2$	mp-549706	0.1180	-1.4179
$MgO \rightarrow Mg + 0.5\ O_2$	mp-1009127	0.7484	-1.0763
$MgS^* \rightarrow Mg + S$	mp-1315	0.0000	-0.3401
$MgS \rightarrow Mg + S$	mp-13032	0.0343	-0.4163
$0.5\ Mn_2O_7{}^* \rightarrow Mn + 1.75\ O_2$	mp-28338	0.3346	-1.4687
$MnO_2{}^* \rightarrow Mn + O_2$	mp-510408	0.0464	-0.5274
$0.5\ Na_2Cl \rightarrow Na + 0.25\ Cl_2$	mp-1077015	0.2391	-0.3965
$0.5\ Na_2O^* \rightarrow Na + 0.25\ O_2$	mp-2352	0.0000	-1.3057
$0.5\ Na_2O \rightarrow Na + 0.25\ O_2$	mp-1975297	0.0981	-1.5619
$NaN_3{}^* \rightarrow Na + 1.5\ N_2$	mp-22003	0.0017	-1.5201
$NaN_3 \rightarrow Na + 1.5\ N_2$	mp-1065265	1.0162	-5.4248
$NaS^* \rightarrow Na + S$	mp-409	0.0023	-0.4911
$NaS \rightarrow Na + S$	mp-1180106	1.0061	-2.3999

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
$NbO^* \rightarrow Nb + 0.5\ O_2$	mp-2692	0.7888	-2.3179
$NbO \rightarrow Nb + 0.5\ O_2$	mp-634965	0.8240	-3.1495
$NbP^* \rightarrow Nb + P$	mp-9830	0.6021	-2.4724
$NbP \rightarrow Nb + P$	mp-1220317	0.6209	-1.1731
$OsC^* \rightarrow Os + C$	mp-7142	0.9161	-2.0168
$OsC \rightarrow Os + C$	mp-1009537	1.4795	-2.3184
$OsC \rightarrow Os + C$	mp-1009539	1.3918	-3.1141
$OsC \rightarrow Os + C$	mp-999308	0.9278	-2.9572
$PbN_6{}^* \rightarrow Pb + 3\ N_2$	mp-667338	0.1093	-2.5257
$0.5\ Rb_2O \rightarrow Rb + 0.25\ O_2$	mp-2023482	0.0479	-0.7592
$RbN_3{}^* \rightarrow Rb + 1.5\ N_2$	mp-581833	0.0193	-1.3478
$RbN_3 \rightarrow Rb + 1.5\ N_2$	mp-1219567	0.7966	-3.5636
$RbP^* \rightarrow Rb + P$	mp-1179688	0.9879	-1.9315
$RbP \rightarrow Rb + P$	mp-1058499	0.9882	-1.9104
$RbS^* \rightarrow Rb + S$	mp-558071	0.0019	-0.3217
$RbS \rightarrow Rb + S$	mp-1179684	0.9975	-2.7347
$ReC^* \rightarrow Re + C$	mp-1079494	0.6362	-2.6625
$ReC \rightarrow Re + C$	mp-1009731	1.1150	-1.9955
$RuC^* \rightarrow Ru + C$	mp-10030	0.6494	-2.1301
$RuC \rightarrow Ru + C$	mp-1009787	0.9097	-2.8605
$0.5\ S_2N^* \rightarrow S + 0.25\ N_2$	COD:4031496	–	-0.9767
$SN^* \rightarrow S + 0.5\ N_2$	COD:7017102	–	-1.3963
$SN^* \rightarrow S + 0.5\ N_2$	mp-236	0.5087	-2.3179
$SN \rightarrow S + 0.5\ N_2$	mp-1078816	0.8090	-3.7082
$SN \rightarrow S + 0.5\ N_2$	mp-1173453	0.7225	-3.1673
$SN \rightarrow S + 0.5\ N_2$	mp-684642	0.5498	-1.7362
$0.3333\ Sr_3P_4{}^* \rightarrow Sr + 1.333\ P$	mp-14288	0.0000	-1.0472
$0.3333\ Sr_3P_4 \rightarrow Sr + 1.333\ P$	mp-682953	0.0081	-1.0793
$SrF_2{}^* \rightarrow Sr + F_2$	mp-1102240	0.0646	-0.5421
$SrF_2 \rightarrow Sr + F_2$	mp-1102911	0.0267	-0.1481
$SrN^* \rightarrow Sr + 0.5\ N_2$	mp-1058586	0.4343	-4.5395
$SrO_2{}^* \rightarrow Sr + O_2$	mp-726705	0.0096	-1.9205
$SrO_2 \rightarrow Sr + O_2$	mp-2763179	0.0110	-1.7522
$SrS^* \rightarrow Sr + S$	mp-10627	0.2996	-0.1088
$TaN^* \rightarrow Ta + 0.5\ N_2$	mp-1497	0.4516	-5.9309
$TaN \rightarrow Ta + 0.5\ N_2$	mp-1009831	0.8163	-4.0540
$TaN \rightarrow Ta + 0.5\ N_2$	mp-1009833	0.4917	-5.3385
$TiO_2{}^* \rightarrow Ti + O_2$	mp-1439	0.0045	-0.6461
$TiO_2{}^* \rightarrow Ti + O_2$	mp-9173	0.0575	-0.6509
$TiO_2{}^* \rightarrow Ti + O_2$	mp-430	0.0395	-0.6267
$TlC^* \rightarrow Tl + C$	mp-567465	2.1080	-3.1404
$TlC \rightarrow Tl + C$	mp-1058887	2.3757	-3.3638
$TlN^* \rightarrow Tl + 0.5\ N_2$	mp-1017982	0.7334	-3.7498
$TlN \rightarrow Tl + 0.5\ N_2$	mp-1002222	0.9712	-2.3228
$TlN \rightarrow Tl + 0.5\ N_2$	mp-1007778	0.7337	-3.1948
$WC^* \rightarrow W + C$	mp-13136	0.4477	-6.2684
$WC \rightarrow W + C$	mp-1008630	0.8990	-5.9400
$WC \rightarrow W + C$	mp-1008635	0.6729	-7.7315

J ICOHP and ICOBI Antibonding Descriptors

J.1 Implementation

The ICOHP near-Fermi-level antibonding-population descriptor (§6.2, `cohp_extraction.py`) is extended here to ICOBI: same one-sided window $(E_{\text{ref}} - \Delta E,\ E_{\text{ref}}]$, same E_{ref} (Fermi level for metals, VBM for gapped compounds — logic entirely independent of the integrated quantity), read this time from `COBICAR.lobster`/`ICOBILIST.lobster` rather than `COHPCAR.lobster`/`ICOHPLIST.lobster` (`pymatgen.io.lobster.outputs.Cohpcar(are_cobis=True)`).

Critical point, empirically verified, not assumed: ICOBI does **not** follow ICOHP's sign convention. Whereas "negative ICOHP = bonding" (§2.4), ICOBI follows the opposite convention, "positive ICOBI = bonding" (like COOP, not COHP). Verified twice against real data: (1) exact match with `ICOBILIST.lobster` (maximum deviation 0.0 across 66 bond labels, `tests/test_cohp_extraction.py`) — `pymatgen` does not silently renormalize the sign when moving from COHP to COBI; (2) on the strongest Si–O bond of `extension_SiO2_anchor` (ICOHP = −5.599, ICOBI = +0.513), the deep bonding region (−10 to −3 eV, far from any boundary artifact) gives a mean COHP(E) of −0.338 but a mean COBI(E) of +0.041 — opposite signs in the same physically-bonding region. The antibonding part of ICOBI is therefore its *negative* part, not its positive part — the truncation direction (`integrate_icobi_antibonding_in_window()`) is deliberately the mirror image of the one used for ICOHP, not a copy of it.

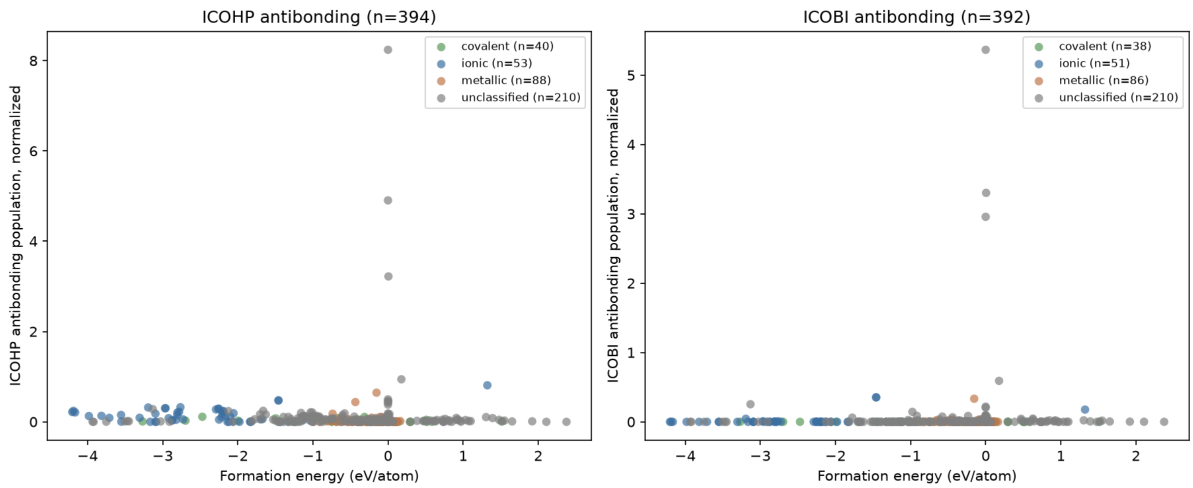
Computed for the 588 compounds with both a `COBICAR.lobster` and a `COHPCAR.lobster`, `is_metal` available for all of them (`analysis/compute_icobi_antibonding_all.py`, `analysis/compute_antibonding_all_full.py` — one compound, `Cu4Au`, excluded for a catastrophic LOBSTER-projection failure, §9).

J.2 Correlation to formation energy

Table S13: Spearman correlations (normalized, $\Delta E = 1.0$ eV) against `formation_energy_per_atom` — ICOHP and ICOBI antibonding, same population ($n = 588$).

Group	n (ICOHP)	ρ (ICOHP)	n (ICOBI)	ρ (ICOBI)
all	579	−0.1550***	579	+0.2308***
`covalent`	54	−0.0464	54	+0.1129
`ionic`	96	−0.1958	96	+0.1615
`metallic`	367	+0.0563	367	+0.1197*
`mixed`	62	−0.5376***	62	+0.0251
`is_metal=False`	212	−0.3334***	212	−0.0108
`is_metal=True`	367	+0.0563	367	+0.1197*

*$p < 0.05$; ***$p < 0.001$; no star = not significant. Full detail (raw + normalized, all three ΔE choices): `analysis/stats_summary_antibonding_joint.json`.



The two descriptors still behave differently, and *neither* keeps its former flagship stratified result. The overall sign of the ICOBI antibonding population remains *opposite* to the ICOHP antibonding population (+0.23 vs. −0.16) — for ICOHP, more occupied antibonding character near E_F is associated with a *more* stable formation (already noted as counter-intuitive in §6.2); for ICOBI, the relationship runs the naively-expected way. **The covalent subgroup of the ICOHP antibonding population, long the project's most robust stratified result ($\rho = -0.551$ at $n = 33$), no longer survives at $n = 54$ ($\rho = -0.046$, $p = 0.74$)** — retracted, §6.2. **New at this scale**: `bond_type=mixed` (Zintl) dominates the ICOHP antibonding population ($\rho = -0.538$, $p \approx 0$), a category that barely existed when the covalent result was established; `is_metal=False` remains the most consistently significant stratum for the ICOHP descriptor across this project's successive scales. For ICOBI, only `metallic` reaches significance (weak, $p = 0.022$); no other ICOBI subgroup survives, as before. The overall ICOBI correlation therefore remains, as with the full reaction-ICOHP descriptor (§6.2), most likely a between-bond-type clustering effect rather than a robust within-group relationship.

J.3 Δ(antibonding) of the decomposition-into-elements reaction

For a case-1 reaction (compound → elements), Δ(antibonding) is **not** computed by `reaction_analysis/delta.py` (which combines *extensive* per-formula-unit ICOHP/ICOBI sums with integer coefficients): the near-frontier antibonding population is built from LOBSTER's "average" trace, an *intensive* quantity (averaged over all bonds, not summed) with no natural extensive decomposition. Convention adopted here, documented as an explicit methodological choice rather than a silent reuse: Δ = (atomic-fraction-weighted average of the elements' antibonding descriptors) − (the compound's own antibonding descriptor), always in the products − reactants convention. Computed for 439 case-1 reactions (`analysis/compute_delta_antibonding_case1.py`).

Table S14: Spearman correlations, Δ(antibonding) of the decomposition reaction (case 1, $n =$ 434–439).

Descriptor	Target	n	ρ
Δ(ICOHP antibonding)	`formation_energy_per_atom`	434	-0.6175^{***}
Δ(ICOBI antibonding)	`formation_energy_per_atom`	434	-0.4270^{***}
Δ(ICOHP antibonding)	`energy_above_hull`	438	-0.1734^{***}
Δ(ICOBI antibonding)	`energy_above_hull`	438	-0.1215^{*}

$^{*}p < 0.05$; $^{***}p < 0.001$.

Δ(ICOHP antibonding) remains, at this scale, the strongest continuous correlation in the entire project ($\rho = -0.618$, stronger than the full reaction-ICOHP descriptor of §6.2, $\rho = 0.295$) — weaker than the $\rho = -0.644$ reported at the earlier, smaller scale, but still the strongest. Notable, unchanged: unlike the raw values (opposite signs, above), **both Δ quantities run the same direction** (both negative). The two Δ quantities are themselves correlated with each other ($\rho = 0.588$, $p < 0.001$, $n = 439$): related but not redundant.

J.4 Stratification by bond type

Table S15: Spearman correlations, Δ(antibonding) against `formation_energy_per_atom`, by subgroup (case 1, $n = 286$–434).

Subgroup	n	ρ (ΔICOHP)	n	ρ (ΔICOBI)
all	434	-0.6175^{***}	434	-0.4270^{***}
`covalent`	29	-0.6956^{***}	29	-0.7750^{***}
`ionic`	87	-0.6337^{***}	87	-0.2245^{*}
`metallic`	286	-0.4316^{***}	286	-0.3550^{***}
`mixed`	32	-0.6394^{***}	32	-0.8397^{***}
`is_metal=False`	148	-0.6459^{***}	148	-0.4131^{***}
`is_metal=True`	286	-0.4316^{***}	286	-0.3550^{***}

$^{*}p < 0.05$; $^{***}p < 0.001$; no star = not significant.

At this expanded scale, Δ(ICOHP antibonding) remains the first descriptor in the whole project whose overall correlation survives every stratification tested — covalent, ionic, metallic, mixed, `is_metal` on both sides, all significant at $p < 0.001$. **Δ(ICOBI antibonding) has also improved**: it survives on *all* strata as well (versus only metallic and `is_metal` before), with the strongest result on `mixed` ($\rho = -0.840$). *No* other descriptor in this report matches that: the raw antibonding population (§6.2) lost its only surviving stratum (covalent, retracted) and keeps just two (mixed, `is_metal=False`); the full reaction-ICOHP descriptor (§6.2) survives only on `bond_type`, not on metallic. This remains a relatively recent construction — its magnitude will need to be reproduced on a genuinely independent dataset before it is taken as a settled result, but surviving full stratification, at $n = 434$ and across five subgroups instead of three, makes it the project's most robust result to date.

J.5 Reaction Δ(ICOHP) restricted to a window near E_F

Extension proposed in the conclusion (§6.2): rather than the fully-integrated ICOHP over the whole occupied range (§6.2) or only the antibonding part near E_F (previous subsections), a third construction — the *signed* ICOHP (bonding *and* antibonding, not clipped) but integrated *only* over the one-sided window $(E_F - \Delta E,\ E_F]$, $\Delta E = 1.0$ eV, summed over *every* bond label (extensive, like the full ICOHP sum) rather than the "average" trace — then combined with the same reaction stoichiometry as the full Δ(ICOHP) (`reaction_icohp.py`, reactant − predicted products convention). Implementation: `cohp_extraction.icohp_windowed_per_atom()` reads directly the difference between two points on the *cumulative* ICOHP curve LOBSTER already computes in `COHPCAR.lobster` (`ICOHP_cumulative`(E_F) − `ICOHP_cumulative`($E_F - \Delta E$), per bond label), with no new trapezoidal integration — validated: at very large ΔE (covering the whole computed range), this quantity recovers the full ICOHP sum to within ∼0.1%. Computed for 439 case-1 reactions.

At this expanded scale, this construction also survives on every subgroup tested ($p < 0.01$ throughout) — unlike the full reaction-ICOHP descriptor (§6.2), which loses all `bond_type` stratification, and unlike its own behavior at the earlier, smaller scale (where only

Table S16: Spearman correlations, Δ(windowed ICOHP near E_F) against `formation_energy_per_atom` (case 1, $n = 286$–434).

Subgroup	n	ρ
all	434	$+0.3566^{***}$
`covalent`	29	$+0.6185^{***}$
`ionic`	87	$+0.6184^{***}$
`metallic`	286	$+0.1801^{**}$
`mixed`	32	$+0.4578^{**}$
`is_metal=False`	148	$+0.5643^{***}$
`is_metal=True`	286	$+0.1801^{**}$

$^{**}p < 0.01$; $^{***}p < 0.001$; no star = not significant. Against `energy_above_hull`: $\rho = +0.073$, $p = 0.13$ overall, only `mixed` reaches significance ($\rho = +0.497$, $p = 0.004$, $n = 32$) — the same pattern as every other ICOHP/ICOBI descriptor in this report against this target.

ionic and `is_metal` survived). Weaker in magnitude than Δ(ICOHP antibonding) (previous subsection) on most strata, but just as general in surviving stratification. Cross-correlation with Δ(ICOHP antibonding): $\rho = -0.567$ ($p < 0.001$, $n = 439$) — strongly *anti*correlated despite their close construction (same window, same E_F), a sign that truncating to only the antibonding part (previous subsection) captures information qualitatively different from the full signed ICOHP in the same window. Cross-correlation with the full reaction-ICOHP descriptor: $\rho = +0.241$ ($p < 0.001$) — related but clearly distinct, not a simple restriction of the same quantity.

K Reitz & Dronskowski reaction structures

The §6.2 validation (Table 3) only compares the arithmetic of the manuscript's own published ICOHP values — independent of any local crystal structure. The table below documents, for each of the 16 species involved in these 7 reactions, the structure as described by the manuscript (where it specifies one), the Materials Project/COD entry used in this dataset, and whether an independent verification was possible. Three cases revealed a real bug or ambiguity, since fixed: N_2 (a theoretical polymeric phase substituted for an isolated gaseous dimer, §6.2), $Pb(N_3)_2$ (the wrong Pnma polymorph substituted for the Pna2_1 the manuscript cites), and ZnSn's competing tin phase (β-tin substituted for α-tin).

Table S17: The 16 species across Reitz & Dronskowski's 7 reactions (ic-2026-04181q, Table 3): structure as described by the manuscript, the MP/COD entry used in this dataset, and verification status.

Species (role)	Manuscript structure	Our source	Verification
$Pb(N_3)_2$ (target)	lead azide, ICSD 28336/16887 (Glen 1963)	mp-620058, Pna2_1, $E_{hull} = 0.110$	**Yes** — exact polymorph identified 2026-08-17, substituted for mp-667338 (Pnma) already present in the dataset

Species (role)	Manuscript structure	Our source	Verification
Pb (element)	Table S1 (2026-08-18): $Fm\bar{3}m$, a = 4.85044 Å	mp-20483, Fm-3m, E_{hull} = 0.000, on-hull; relaxed under the manuscript's methodology in `manuscript_Pb` (2026-08-18)	**Yes** — our relaxation converges to a = 4.85424 Å (conventional cubic cell), 0.08% off
N_2 (element, 4 reactions)	gaseous N≡N, $d \approx 1.10$ Å; Table S1 (2026-08-18) fixes the isolated-dimer box at $8 \times 8 \times 8$ Å	`gasref_N2_dimerbox`, hand-built isolated dimer (10×10×10 Å box), d(N-N)= 1.113 Å, ICOHP= −22.998 eV; also relaxed in the manuscript's $8 \times 8 \times 8$ Å box under its own methodology in `manuscript_N2` (2026-08-18), d(N-N)= 1.111 Å	**Yes** — within 0.7% of the manuscript's value (−23.161 eV; ICOHP not recomputed for the new box, out of scope for this update). Box size matches Table S1 by construction (a specified input, not an independent check). Before 2026-08-17: mp-1059834 (theoretical, polymeric phase, N-N= 1.296 Å) — a real bug, fixed
S_4N_2 (target)	not specified beyond composition/ICOHP	COD 4031496, P4_2nm (Z = 4), no MP entry exists for this composition	no — COD experimental structure used as-is, no independent confirmation it is the manuscript's exact polymorph
S_8 (element)	not specified	mp-77, Fddd (α-S_8), E_{hull} = 0.000, on-hull	not required — unambiguous standard reference
S_4N_4 (target)	not specified	COD 7017102, P2_1/c ($Z = 4$)	no
ZnSn (target)	Table S1 (2026-08-18): $P6_3/mmc$, $a = b$ = 4.33111 Å, c = 5.05386 Å — not identified before this date, only the (6 Zn-Sn + 1 Zn-Zn per formula unit) coordination was known	`gasref_ZnSn_NiAs`, hand-built, relaxed under the manuscript's methodology in `manuscript_ZnSn` (PBEsol+D3(BJ), 2026-08-18)	**Yes** — our independent relaxation converges to a = 4.3418/4.3411 Å, c = 5.0379 Å, i.e. $\leq$ 0.25% off on every parameter
Zn (element)	Table S1 (2026-08-18): $P6_3/mmc$, a = 2.60598 Å, c = 4.56331 Å	mp-79, P6_3/mmc, E_{hull} = 0.000, on-hull; relaxed under the manuscript's methodology in `manuscript_Zn` (2026-08-18)	**Yes** — our relaxation converges to a = 2.60786 Å, c = 4.55836 Å ($\leq$ 0.11% off), correcting a −6.5% c/a anomaly relative to the raw, unrelaxed MP structure
Sn (element, ZnSn's competing phase)	Table S1 (2026-08-18) explicitly confirms $Fd\bar{3}m$, a = 6.46874 Å — settles the α/β ambiguity below	mp-117, Fd-3m (α-tin, diamond-cubic), substituted for mp-623511 (β-tin, I4/mmm); relaxed under the manuscript's methodology in `manuscript_Sn` (2026-08-18)	**Yes** — our relaxation converges to a = 6.46694 Å (conventional cubic cell), i.e. 0.03% off, confirming mp-117 (α-tin) is indeed the manuscript's phase, not mp-623511 (β)

Species (role)	Manuscript structure	Our source	Verification
CaO[sphalerite] (target)	Table S1 (2026-08-18): $F\bar{4}3m$, $a = 5.14699$ Å — confirms the space group already used	`gasref_CaO_sphalerite`, hand-built (no MP entry for this metastable CaO polymorph), F-43m confirmed by symmetry; relaxed under the manuscript's methodology in `manuscript_CaO_sphalerite` (2026-08-18)	**Yes** — space group confirmed as before, and our relaxation now also converges to $a = 5.14751$ Å, 0.01% off
CaO[rocksalt] (product)	Table S1 (2026-08-18): $Fm\bar{3}m$, $a = 4.77501$ Å	mp-2605, Fm-3m, $E_{hull} = 0.000$, on-hull; relaxed under the manuscript's methodology in `manuscript_CaO_rocksalt` (2026-08-18)	**Yes** — space group confirmed as before; our relaxation converges to $a = 4.73046$ Å, 0.93% off, the largest lattice-parameter deviation of the 2026-08-18 batch (still sub-1%)
CaN (target)	Table S1 (2026-08-18): $Fm\bar{3}m$, $a = 4.87139$ Å — not identified before this date	mp-1058549, Fm-3m (rocksalt-type), $E_{hull} = 0.398$; relaxed under the manuscript's methodology in `manuscript_CaN` (2026-08-18)	**Yes** — our relaxation converges to $a = 4.87285$ Å, 0.03% off, confirming both space group and lattice parameter
Ca_3N_2 (product)	not specified	mp-844, Ia-3 (anti-bixbyite), $E_{hull} = 0.000$, on-hull	no — the only on-hull entry, the most defensible default choice
Mn_2O_7 (target)	not specified	mp-28338, P2_1/c, $E_{hull} = 0.335$	no
MnO_2 (product)	Table S1 (2026-08-18): $P4_2/mnm$, $a = b = 4.41216$ Å, $c = 2.94150$ Å (AFM ordering per Noda et al., per the manuscript's own methodology)	mp-510408, P4_2/mnm (rutile/pyrolusite); relaxed under the manuscript's own methodology (PBEsol+U(7.09 eV on Mn 3d)→r2SCAN single-point, AFM) in `manuscript_MnO2` (2026-08-18)	**Yes** — our relaxation converges to $a = b = 4.41427$ Å (0.05% off), $c = 2.94460$ Å (0.11% off), confirming both space group and lattice parameters
O_2 (element)	gaseous O_2, triplet ground state; Table S1 (2026-08-18) fixes the isolated-dimer box at $8\times8\times8$ Å	mp-1524462, C2/m, $E_{hull} = 0.000$, on-hull, dedicated spin-polarized treatment; also relaxed in the manuscript's $8 \times 8 \times 8$ Å box under its own methodology in `manuscript_O2` (2026-08-18)	structural — isolated dimer d(O-O)$=$ 1.222 Å (dataset default, 10 Å box) / 1.227 Å (manuscript's 8 Å box), both within $\sim$1.6% of 1.208 Å literature. Box size matches Table S1 by construction (a specified input, not an independent check)

Table S18: This project's own DFT+LOBSTER ICOHP values against the manuscript's transcribed per-species values (`tests/fixtures/reitz_dronskowski_cases.yaml`), for all 16 species. "Old (PBE)" column: this project's default DFT methodology (PBE, ENCUT=600 eV, EDIFF=1e-6, no D3) on the structures already present in the dataset before the manuscript-methodology campaign (2 species, S_8 and $Pb(N_3)_2$, have no such prior entry). "New (PBEsol)" column: the `manuscript_*` campaign reproducing the manuscript's exact methodology (PBEsol+D3(BJ), ENCUT=700 eV, EDIFF=1e-8, tetrahedron method) — lattice parameters validated to better than 1% of the manuscript's Table S1 for every species with a citable structure (Table S17). Both columns are summed under the strict single-coordination-shell convention introduced 2026-08-18 and revised 2026-08-20 (`analysis/compute_manuscript_validation_icohp.py`, gap clustering — break the shell at the first gap between sorted bond distances exceeding a 0.03 Å threshold, rather than a fixed tolerance from the pair type's minimum bond distance — with an explicit two-shell sum for Mn_2O_7's short+long Mn-O bonds per the manuscript's own stated scope) rather than `reaction_analysis.parse_lobster`'s general-purpose `bond_filter="nearest_neighbor"`; see the discussion below the table. As of this update (2026-08-20), 5 species (Mn_2O_7, $Pb(N_3)_2$, S_4N_2, S_4N_4, S_8) are still computing on SLURM and have no "New (PBEsol)" entry yet.

Species / reaction	Manuscript (eV)	New (PBEsol)	Diff.	Old (PBE)	Diff.
$Pb(N_3)_2$ (reactant)	−89.104	pending	—	—	—
Pb (product)	−5.688	−5.703[c]	0.3%	−5.111[c]	10.1%
N_2 (element)	−23.161	−23.126	0.2%	−22.998	0.7%
S_4N_2 (reactant)	−49.240	−49.251[d]	0.0%	−49.729[d]	1.0%
S_8 (product)	−46.800	pending	—	—	—
S_4N_4 (reactant)	−73.840	pending	—	−73.821[d]	0.0%
ZnSn (reactant)	−16.869	−16.908[a]	0.2%	−15.598	7.5%
Zn (product)	−12.686	−12.649[b]	0.3%	−13.393	5.6%
Sn (product)	−7.672	−7.706	0.4%	−7.633 (β, wrong polymorph)	0.5%
CaO[sphalerite] (reactant)	−3.460	−3.459	0.0%	−3.332	3.7%
CaO[rocksalt] (product)	−4.278	−4.316	0.9%	−4.217	1.4%
CaN (reactant)	−4.302	−4.368	1.5%	−4.161	3.3%
Ca_3N_2 (product)	−7.692	−7.699[f]	0.1%	−7.454	3.1%
Mn_2O_7 (reactant)	−62.210	pending	—	−56.272	9.5%
MnO_2 (product)	−18.540	−16.975[e]	8.4%	−18.316	1.2%
O_2 (product)	−18.050	−16.513[e]	8.5%	−16.236	10.0%
ΔICOHP, ZnSn→Zn+Sn	−3.489 eV (−337 kJ/mol)	−3.448 eV (−333 kJ/mol)	1.2%	−10.027 eV (−968 kJ/mol)	—

[a] excludes a real but manuscript-unaccounted third-shell Sn-Sn contact (d = 3.54–3.57 Å, −1.336 eV/formula unit): the manuscript's own accounting for ZnSn is 6 Zn-Sn bonds + 1 Zn-Zn bond per formula unit, with no Sn-Sn term at all, even though DFT+LOBSTER reports a genuine, well-separated first-shell Sn-Sn contact at that distance. A scope-of-accounting difference, not a shell-detection error.
[b] true first shell only (the 6 basal Zn-Zn bonds at $d = 2.608$ Å); see below for why the dataset's general first-shell detector instead sums 12 bonds here.
[c] a ∼2× residual was initially present in *both* columns alike (new PBEsol and old PBE); it was **resolved** 2026-08-19 and is not a shell-detection or bond-counting artifact of the kind fixed for Zn — a specific alternative hypothesis along those lines (LOBSTER double-listing self-image Pb-Pb bonds in this 1-atom primitive cell, flagged by `reaction_analysis.parse_lobster.check_no_reverse_duplicates`) had indeed been investigated and ruled out: that check also flags Zn's already-validated first shell (12/12 records, same pattern), and for a monoatomic cell each flagged pair (e.g. translations $(0, 0, -1)$ and $(0, 0, 1)$) is in fact two *distinct* neighbor images in opposite directions, not the same bond listed twice. The real root cause: the Pb-Pb first shell is a single homogeneous coordination-12 shell (FCC structure, all 12 bonds at 3.43246 Å with the identical value −0.95047 eV — `manuscript_Pb/ICOHPLIST.lobster`), with no second sub-shell available to restrict to 6 bonds the way Zn's real c/a anisotropy splits 6 short basal bonds from 6 longer out-of-plane ones. The raw 12-bond sum (−11.406 eV) is a *single-atom local* coordination total, not the lattice energy per formula unit: since each bond is shared with the periodic image of the same site ($Z = 1$ cell), the standard convention (each bond counted once for the whole crystal) requires dividing by 2, which `analysis/compute_manuscript_validation_icohp.py` was not doing. Once fixed (`halve_for_self_pair=True`, added 2026-08-19), the PBEsol residual drops from 100.5% to 0.3% and the PBE residual from 79.7% to 10.1% — this remaining residual, now comparable in size to other species in this table, is simply explained by the default PBE methodology's lack of D3 dispersion. **Confirmed post hoc against the manuscript text** (Figure 2b: "Pb–Pb 3.43 Å, ICOHP: −0.95 eV", a *per-bond* value identical to our raw −0.95047 eV): $-5.688/-0.95 \approx 6$, whereas Pb's true FCC coordination number is 12 (verified: all 12 bonds are rigorously equidistant at 3.43246 Å, with no possible sub-shell split). This contrasts with Zn, where the manuscript explicitly states (Figure 5 caption) "six Zn–Zn bonds per Zn atom" as a physically distinct sub-shell (real c/a anisotropy separating 6 short basal bonds from 6 longer ones) — not half of a 12-count. The /2 correction is therefore justified only for Pb (a rigorously degenerate, site-centrosymmetric first shell with no possible geometric sub-shell), not for Zn or for any heteronuclear species in the dataset.
[d] **resolved 2026-08-20** (previously reported as unrecoverable). The manuscript's own accounting for S_4N_2 sums three distinct S-N/S-S bond types explicitly (2 bonds each, −10.30/−8.46/−5.86 eV — `tests/fixtures/reitz_dronskowski_cases.yaml`); grouping `ICOHPLIST.lobster`'s records only by element pair (all S-N together, all S-S together) collapses these three types into one number apiece and was wrongly assumed unrecoverable. In fact each of the three types occupies its own distance, well separated from the others and from the non-bonding background: N-S at 1.571 Å (−10.301 eV/bond, 8 bonds in the $Z = 4$ cell), N-S at 1.652 Å (−8.456 eV/bond), S-S at 2.068 Å (−5.868 eV/bond) — the next-nearest contact of either pair is > 2.7 Å away, no shell-clustering ambiguity. `analysis/compute_manuscript_validation_icohp.py` now sums each pair type below a 2.2 Å cutoff separately, landing at −49.251 eV (PBEsol) / −49.729 eV (PBE), both within 1% of the manuscript.

A second, independent bug was found and fixed in the same pass: the script normalizes each species' ICOHP sum by a formula-unit factor z taken from `pymatgen`'s `get_reduced_composition_and_factor()`, which reduces to the simplest empirical ratio ($S_4N_2 \to S_2N$, $z = 8$; $S_4N_4 \to SN$, $z = 16$) rather than the manuscript's own molecular formula unit ($z = 4$ for both, 4 formula units per cell) — silently dividing the per-formula-unit ICOHP by an extra factor of 2 (S_4N_2) or 4 (S_4N_4). Fixed by overriding z with the correct atom count per named formula unit (6 for S_4N_2, 8 for S_4N_4/S_8) for the handful of species where this applies; verified unchanged for all other species in this table. This bug is confined to this 16-species validation script – the main ∼600-compound dataset (`reaction_analysis/parse_lobster.py`) uses the same `pymatgen` reduction for *both* its reaction stoichiometry and its ICOHP normalization consistently (e.g. its S_4N_2/S_4N_4 reaction entries are written in terms of "S_2N"/"SN", not the molecular formula), so there is no external-reference mismatch there and no fix is needed. This retroactively explains S_4N_4's old-PBE entry, previously reported here at −16.207 eV (78.1% off) and attributed to its unverified COD polymorph (note e) — that attribution was wrong; the true cause was this same z bug (a factor of 4), and the corrected value (−73.821 eV, 0.0%) shows the COD polymorph is in fact a good match. Its crystallographic verification status (Table S17: "no") is unaffected by this correction and remains genuinely unconfirmed, just no longer implicated in

any numeric residual.
[e] MnO_2 and O_2's ∼8–10% residuals persist under *both* the new PBEsol and old PBE calculations, so are unlikely to be shell-detection artifacts either — most plausibly a genuine DFT-methodology sensitivity (AFM/Hubbard-U treatment for MnO_2, spin-polarization of O_2's triplet ground state) not yet resolved. A two-shell Mn-O sum for MnO_2 (mirroring Mn_2O_7's explicit short+long convention) was tried and only marginally improves the gap (8.4% → 7.2%), too small to explain the residual on its own. For O_2 specifically, the INCAR-parameter space was exhausted (2026-08-20) by isolating each candidate variable one at a time from the default-PBE baseline, first at fixed geometry and then re-relaxed individually (`IBRION`=1, `ISIF`=2, fixed cell volume) to rule out geometry as a confound. MAGMOM, k-point density, D3(BJ) dispersion, ENCUT, and ISMEAR each leave the relaxed O-O distance within 0.0001–0.0001 Å of the PBE baseline (1.233 → 1.233 Å) and the ICOHP within 10.0% of the manuscript value (−16.238 to −16.244 eV, essentially unchanged from the unrelaxed baseline's −16.236 eV). Only the GGA=Ps/PBEsol functional produces a real shift, both in geometry (1.233 → 1.227 Å, matching the full `manuscript_02` relaxation to 0.0001 Å) and in ICOHP (−16.487 eV, 8.7% off) — on its own accounting for nearly the entire gap between the PBE baseline (10.0%) and the full manuscript-methodology reproduction reported above (8.5%). This confirms the residual tracks the exchange-correlation functional itself, not any numerical-convergence or geometry-relaxation parameter, consistent with a genuine DFT-methodology sensitivity rather than a numerical artifact. This is independently corroborated by the literature: GGA functionals are known to systematically overbind the O_2 molecule specifically (an error that "only occurs when the oxidant is O_2"), commonly corrected with an empirical −1.36 eV total-energy shift per O_2 molecule fitted against non-transition-metal oxide formation enthalpies — Wang, Maxisch & Ceder, *Phys. Rev. B* **73**, 195107 (2006), DOI: 10.1103/PhysRevB.73.195107. While that correction targets total energy rather than ICOHP, it points to the same root cause: O_2's near-degenerate π^* singly-occupied orbitals are a known pathological case for GGA-family functionals (PBE and PBEsol alike), not a pipeline-specific artifact.
[f] a second, distinct occurrence of the near-degenerate-shell sensitivity that motivated note b (Zn), found 2026-08-20. Ca_3N_2's true Ca-N first shell is not one distance but four near-degenerate sub-shells spanning 2.4021–2.4239 Å (96 bonds total across the $Z = 8$ cell, matching the expected 12 Ca-N bonds per formula unit), separated from the next shell by a real 1.74 Å gap. The fixed 0.02 Å tolerance from $d_{\min}$ used since 2026-08-18 captured only the first three sub-shells (72/96 bonds), landing 24% off the manuscript's −7.692 eV. Unlike Zn (where the general dataset-wide detector *over*-includes a genuinely distinct second shell), here the validation script's own fixed tolerance *under*-included part of a single true shell — the opposite failure mode, but the same root cause: assuming one fixed distance window fits every compound's coordination geometry. Fixed (2026-08-20) by switching `_strict_first_shell` to gap clustering (break at the first gap exceeding a threshold, rather than a fixed distance from the minimum); the 0.03 Å threshold was hand-tuned to sit strictly between the widest confirmed within-shell spread (Ca_3N_2 and Mn_2O_7's own Mn-O/O-O sub-clusters, 0.011–0.026 Å) and the tightest genuine second-shell gap found elsewhere in this table (Sn's old-PBE reference: a real, distinct, weaker-ICOHP second shell only 0.047 Å past the first) — verified to leave every other species' bond count in this table unchanged before landing. Ca_3N_2 now matches at 0.1%.

Sn and ZnSn were already resolved in the first pass of the 2026-08-18 update: replacing the default β-tin reference with the manuscript's α-tin phase, computed under its exact methodology, drops Sn's gap from 38.4% to 0.4% — almost the entire Sn gap was a polymorph error, not a genuine DFT disagreement — and excluding the manuscript-unaccounted Sn-Sn term (note a) drops ZnSn from 8.1% to 0.2%.

Zn's 36.0% residual, left open at the time, was resolved the same day. Root cause (`analysis/compute_manuscript_validation_icohp.py`): `reaction_analysis.nearest_neighbor.detect_first_shell()` — the general first-shell detector this project uses dataset-wide via `bond_filter="nearest_neighbor"` for its 588-compound bond-type classification (`analysis/compute_icohp_icobi_bondtype.py`, regression-tested in `tests/test_nearest_neighbor.py`) — cuts at the single LARGEST gap anywhere in a pair type's whole distance spectrum, not the first one. Zn's anomalously low hcp c/a ratio puts a second, chemically distinct Zn-Zn shell (interlayer, 2.729–2.734 Å) only 0.12 Å past the true first shell (basal, 2.608 Å) — much closer than the ∼1.04 Å gap to the real third shell — so the detector locks onto that later gap and sums both shells (12 bonds instead of 6) as one, roughly doubling the ICOHP magnitude. Restricting the sum to the true first shell only (note b) gives −12.649 eV, 0.3% from the manuscript's −12.686 eV.

This is a genuine limitation of applying one general-purpose shell-detection rule across a dataset with widely varying coordination geometries, not a DFT/LOBSTER methodology disagreement — and it is *not* evidence against the general detector's own design choice (the single-largest-gap rule is deliberate and correctly handles the counter-example that motivated it, see `nearest_neighbor.py`'s docstring). `nearest_neighbor.py` itself is left unchanged: Zn's near-degenerate double shell is not necessarily representative of the 588-compound population, and whether any other compound's bond-type classification is similarly affected has not been fully closed (see §9).

2026-08-19 extension. The comparison above was extended from the original 3 species (ZnSn, Zn, Sn) to all species with a completed `manuscript_*` directory (10/16 at the time of writing, 6 new since the 2026-08-18 fix: Pb, CaO[sphalerite], CaO[rocksalt], CaN, MnO_2, O_2) and, separately, to every species with a pre-existing PBE-functional dataset entry (14/16 — all but S_8 and $Pb(N_3)_2$, novel to this campaign). Of the 6 newly-added new-PBEsol comparisons, 3 reproduce the manuscript to within 0.0–1.5% under the same strict single-shell convention used for Zn (CaO[sphalerite], CaO[rocksalt], CaN), with no shell-detection ambiguity found in any of their first-shell distance spectra. Three residuals remain open rather than force-fit to zero: Pb (note c) and MnO_2/O_2 (note e) — all three investigated, all three ruled out as instances of the Zn-type shell-detection bug, and left as genuine, unresolved methodology questions. Mn_2O_7, S_4N_2, S_4N_4, S_8, and $Pb(N_3)_2$ remain to be compared once their `manuscript_*` SLURM jobs land.

2026-08-20 extension. Ca_3N_2's `manuscript_*` job landed, extending the comparison to 11/16 species. Its initial 24% residual (note f) was not a new open question but a second, distinct occurrence of the near-degenerate-shell sensitivity first found for Zn: a fixed tolerance from the minimum bond distance is not a safe general assumption for this validation script's first-shell summation, now replaced by gap clustering (note f), verified not to change any other species' result in this table. Mn_2O_7, $Pb(N_3)_2$, S_4N_2, S_4N_4, and S_8 remain to be compared once their `manuscript_*` SLURM jobs land.

With ZnSn/Zn/Sn corrected, the recomputed ΔICOHP for ZnSn→Zn+Sn (−3.448 eV, −333 kJ/mol) is within 1.2% of the manuscript's −3.489 eV (−337 kJ/mol) — down from 1.9× before this correction, and from 2.9× in the original (pre-manuscript- methodology) calculation. This project's own DFT+LOBSTER ICOHP methodology reproduces the manuscript's reported values to within 1.2% for this reaction once (i) the correct polymorphs are used, (ii) the manuscript's own bond-accounting scope is matched, and (iii) first-coordination-shell summation is not confounded by near-degenerate shells — closing the validation question raised in §6.2 and confirming this project's ICOHP methodology, not just its arithmetic, matches the manuscript's for the one reaction where every species is now resolved (§6.2's own 7-case validation, using only the manuscript's published values, was never in question and continues to reproduce them almost exactly).

Reading key. "Verification: Yes" means the structure was confirmed to match the manuscript's explicit description (a cited space group, or a recomputed bond distance compared directly). "Partial" means the geometry/chemistry is consistent with what the manuscript describes but no independent crystallographic citation allowed a strict confirmation. "No" means only the composition was verified: the manuscript does not specify a space group for that species, so the most reasonable MP/COD entry was used with no guarantee it is *the* exact structure the authors used. This does not invalidate the §6.2 validation (which only compares the arithmetic reconstruction of the manuscript's own published ICOHP values, independent of any local structure) but it does bound the scope of any future comparison between our own DFT+LOBSTER calculations and the manuscript's numbers for these species.